

Road to Practitioner: Lecture 0+1

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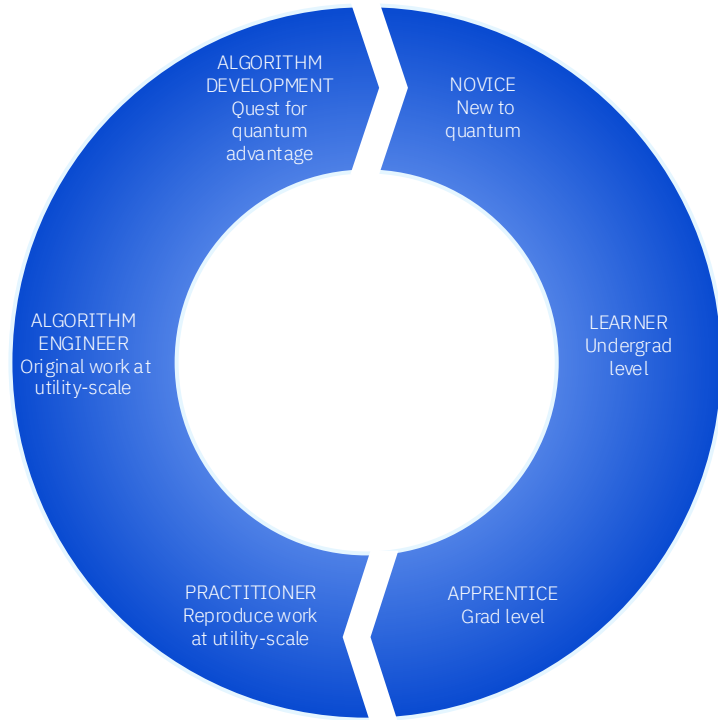
Agenda

1. Overview of Road to Practitioner Program
2. Introduction to Quantum Computing
3. Quantum Computing basics
4. Towards Quantum Advantage
5. Programming Quantum Computers with Qiskit SDK
6. IBM Quantum Platform and learning opportunities
7. Hands-on: Qiskit Installation and first quantum circuit

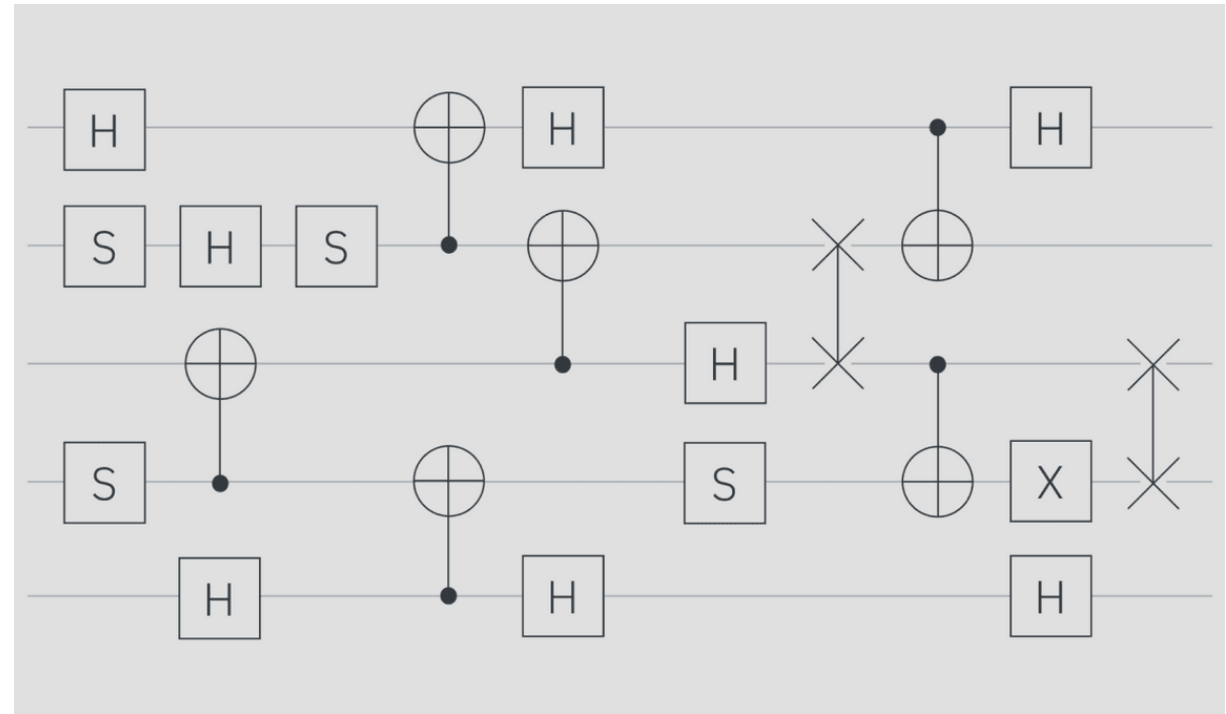
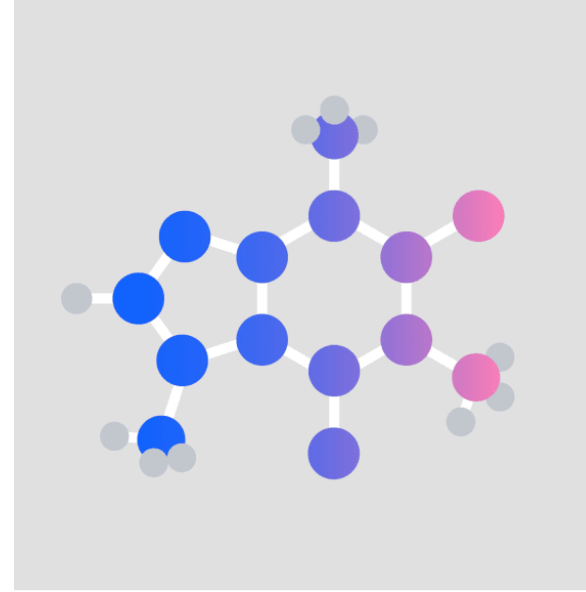


Road to Practitioner (R2P) Program

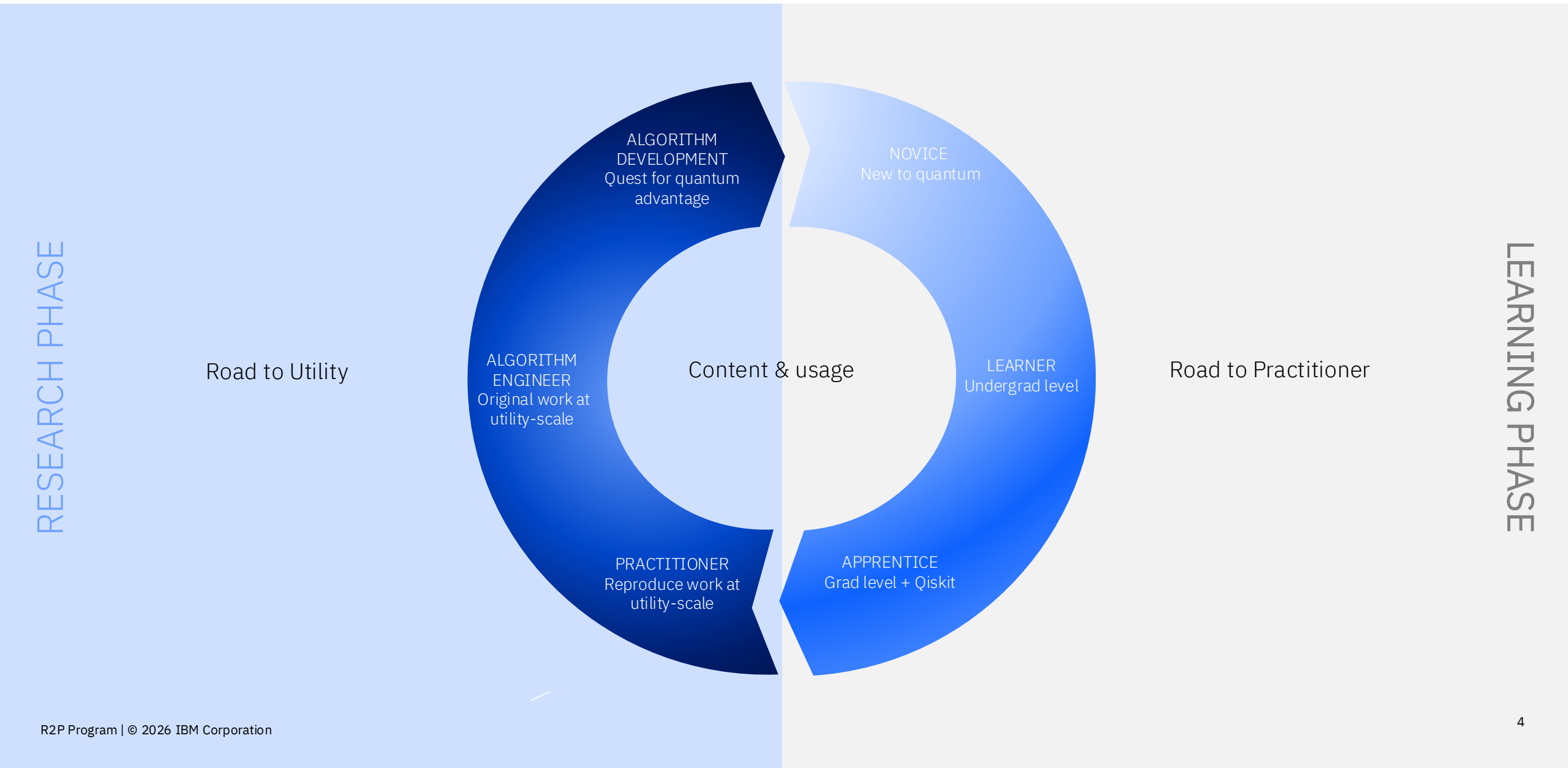
A curated learning journey from novice to practitioner



IBM Quantum

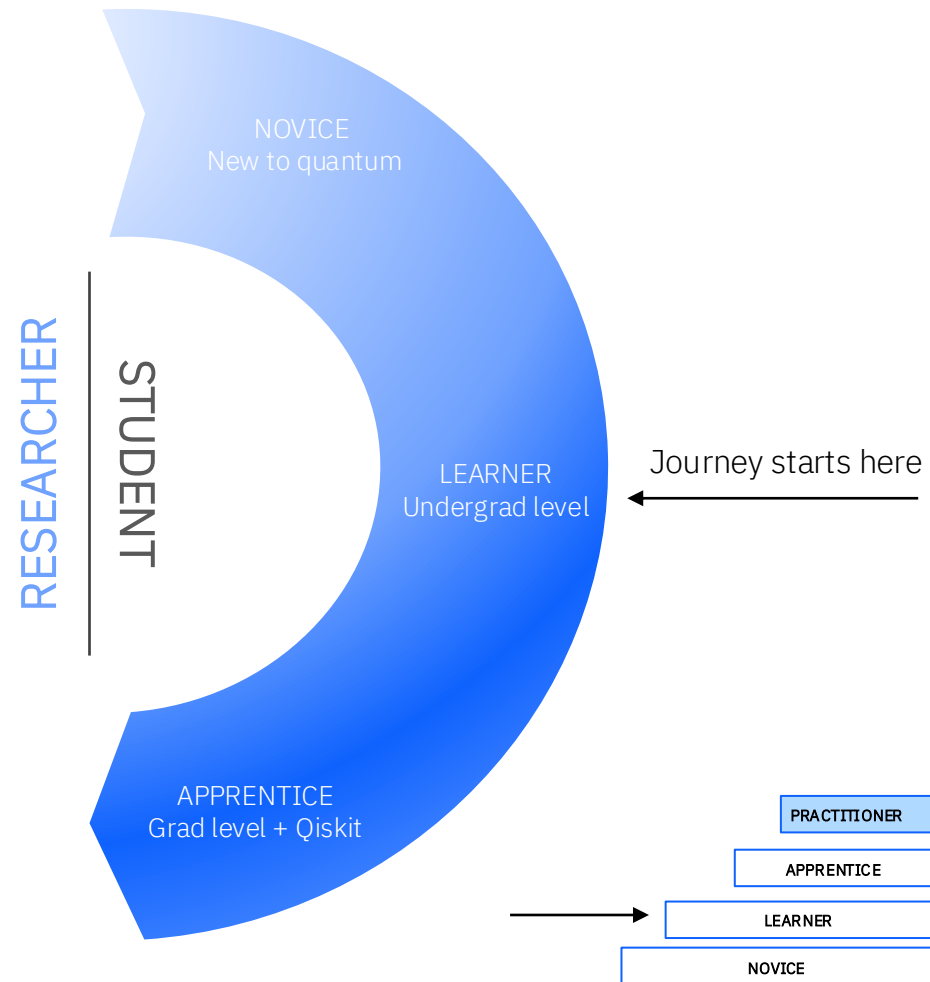


Adoption stages



Road to Practitioner

In a nutshell



12-weeks journey to become practitioner

- **Cohort-Based:** A class of peers encouraging collaborations and peer support and fostering the organic development of local communities.
- **Modular Learning Journey:** Nine lectures spread across six units, integrating theory and hands-on experiences.
- **Hands-On Learning:** Emphasis on practical experience thanks to exercise sessions, code explanations and coding assignments.
- **Capstone Project:** Team-based coding challenges focused on applying on use cases of interest the lessons learned during the program.
- **Complementary Self-Paced Learning:** State-of-the-art digital education content from the IBM Quantum Learning page complement each lecture.
- **Guided Support:** Office hours and dedicated mentorship channels to assist learners throughout the program.
- **Solid foundation for R2U:** A learning journey designed to lay the foundational knowledge needed for the Road to Utility (R2U) program.

Road to Practitioner Curriculum

Road to Practitioner Curriculum		Unit 0: Activation Workshop Theory • R2P program overview • Intro to quantum computing • Towards quantum advantage • Intro to Qiskit SDK • IBM Quantum Platform • IBM Quantum Learning • DEMO Composer Hands-on Qiskit • Install Qiskit and create an IBM account • Run the first circuit: Hello World	Unit 1: Qiskit Basics Theory • The software stack • Algorithm Anatomy: Qiskit Patterns • Qiskit Addons and Functions Hands-on Qiskit • Qiskit Patterns • Qiskit Code Assistant • Scaling the GHZ state to 100 qubits • Encodings and Ansatzes • IQP platform and manage jobs
Unit 2: Quantum Algorithms I Theory • Overview of Variational algorithms: • Case studies: VQE, QAOA • Overview of QML and data encoding methods Hands-on Qiskit • VQE • QAOA • Quantum Kernel Estimation	Unit 3: Quantum Computing Tools and Techniques Theory • Optimizing circuits: Transpilation • Dealing with noise: Overview of different ES and EM techniques (DD, Twirling, REM, ZNE, PEC) • New primitive Samplomatic • Mid-circuits measurements Hands-on Qiskit • Preset and Custom Pass Manager • Comparing results different ES and EM techniques (DD, Twirling, REM, ZNE) • Dynamic Circuits	Unit 4: Quantum Algorithms II Theory • Simulating the dynamic of a Ising model (spin chain + Trotterization) • Sample based methods: SQD/KQD/SKQD • Quantum Centric Supercomputing • Resource management (Slurm) Hands-on Qiskit • Hamiltonian simulation (Ising model with 70 qubits) • Time evolution (Trotterization) • SQD Addon and example	Unit 5: Capstone Project Teamwork Complete a project in cutting edge domains such as: • Quantum diagonalization • Quantum optimization • Quantum simulations Hands-on Qiskit Solve the challenges and run them on hardware. Submission includes: • Jupyter notebook

Road to Practitioner Program

Canada – 1H 2026

Time: 13:00 – 14:30 EST

May

- **Unit 0| Activation workshop + Unit 1| Qiskit Basics**
Lecture 1: Introduction to Qiskit
11th of May
- **Unit 2| Quantum Algorithms I**
Lecture 1: Variational Algorithms
14th of May
- **Unit 2| Quantum Algorithms I**
Lecture 2: QML
21st of May
- **Unit 3| Quantum Computing: Tools and Techniques**
Lecture 1: Transpilation
25th of May
- **Unit 3| Quantum Computing: Tools and Techniques**
Lecture 2: Error Mitigation and Suppression
28th of May

June

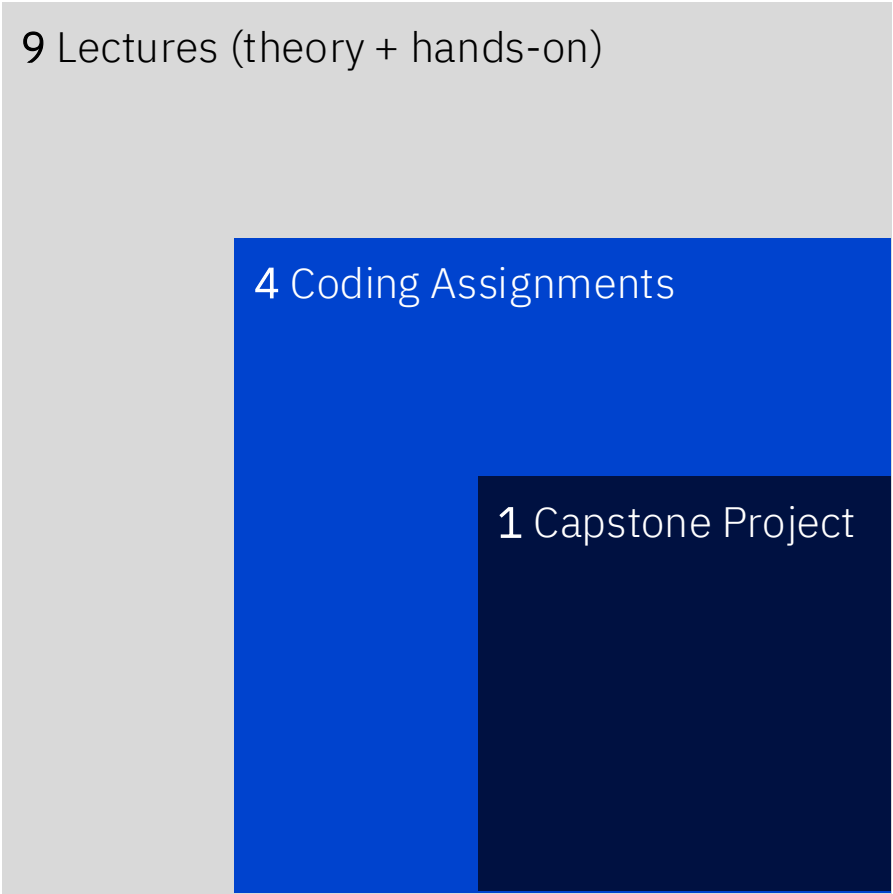
- **Unit 3| Quantum Computing: Tools and Techniques**
Lecture 3: Dynamic Circuits and other Advanced Techniques
4th of June
- **Unit 4| Quantum Algorithms II**
Lecture 2: Hybrid workflows
Sample-based Diagonalization
11th of June
- **Unit 4| Quantum Algorithms II**
Lecture 2: Hamiltonian Simulation
+
Unit 5| Capstone Project: Start
18th of June

July

- **Unit 6| Capstone Project: Submission**
6th of July
- **Unit 6| Capstone Project: Final Ceremony**
9th of July (TBC)

Road to Practitioner

What to expect



Theory



Each lecture has a theory part where the lecturer provides context, motivation, and a deep dive into a specific topic.

Hands-on



Each lecture has a hands-on part where the lecturer will go in more details through some relevant topics covered during the lectures.

Coding Assignments



Coding exercise are assigned at the end of each Unit and meant to test your acquired knowledge and to prepare for the final Capstone Project.

Capstone Project



Team-based work on utility-scale coding challenges, in which teams of 3–5 participants collaboratively develop and submit their solutions, with each submission being graded and requiring a minimum passing score to successfully complete this final phase of the programme.

Road to Practitioner

The Capstone Project

IBM Quantum

Objectives:

- Practice with the latest Qiskit features and tools
- First hands-on experience with algorithms poised for quantum advantage
- Test the acquired knowledge on advanced topics and methodologies

Format:

- Teams of ~3 participants work on coding challenges on the topics covered during the program.
- An auto-grader reviews all submitted solutions and a final ranking is produced.
- Team results are classified as Not Pass, Pass, Good, or Excellent.

Guidelines:

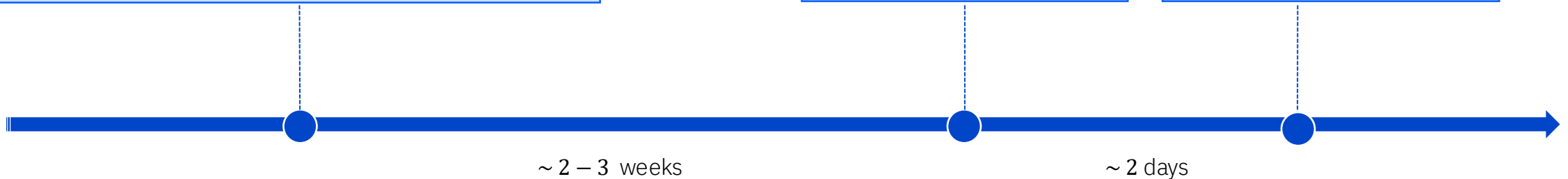
- Team formation (~ 3 people per team).
- Explanation of the challenges.
- Explanation of the submission and evaluation.

Submission:

- Completed notebooks need to be submitted.

Final Ceremony:

- Optional presentations by the teams who provided the best solutions.



Road to Practitioner

Prerequisites

Quantum mechanics basics



- Basics of quantum states and measurements
- Fundamental quantum principles: superposition, entanglement, interference
- Basics of linear algebra: matrix multiplication, etc
- Basics of complex numbers and probability concepts

Recommendations:

- Qiskit YouTube: [What is Quantum Computing?](#)
- [Course: Basics of Quantum information](#)
- Book: Quantum Computation and Quantum Information by Michael A. Nielsen and Isaac L. Chuang – first two chapters
- Book:

Python programming basics



- Variables, data types, and control structures (loops, conditionals)
- NumPy arrays and basic mathematical operations
- Functions and object-oriented programming concepts
- Experience with Jupyter notebooks
- Ability to install and manage Python packages (pip, conda)

Recommendations:

- [Project Jupyter](#)
- [IBM Quantum guide: Online lab environments](#)
- [Visual Studio Code software](#)

IBM Quantum account and Qiskit installation



- Installing Qiskit
- Creating an IBM Quantum account

Recommendations:

- [Install Qiskit](#)
- [Set up your IBM Cloud Account](#)

Course readiness & commitment



- Weekly time commitment: 1.5 hour lectures + coding assignments (variable time) = estimated 3 – 5 hours
- Motivation to complete all assignments and final project/challenge

Who is this program for?



- **Researchers** looking to deepen their expertise in Quantum Computing tools and applications.
- **Students** aiming to make their first steps towards a technical career in quantum computing.
- **Employees** working in technical roles and wanting to learn the latest quantum computing tools.
- **Educators** developing a course or an academic program in quantum computing.
- **Qiskit users** who want to stay current with the latest features and tools.

Who is this program *not* for?



- **Researchers** who have no prior exposure to quantum computing.
- **Students** without foundational knowledge of linear algebra or basic quantum mechanics.
- **Employees** with minimal or no experience in python programming.
- **Educators** interested only in the theoretical aspects of quantum computing.
- **Participants** who cannot dedicate time to the coding assignment and the capstone project.

Road to Practitioner Program

The Badge

- The badge earner must have attended the invite-only Road to Practitioner Program for IBM selected clients.
- The badge earner must have been an active team member and contributor to a Capstone project submission at the end of the Road to Practitioner Program.
- The badge earner must be part of a team that passes the Capstone project at the end of the Road to Practitioner Program.
- The badge earner must individually submit an application form to request the badge.

Road to Practitioner Program

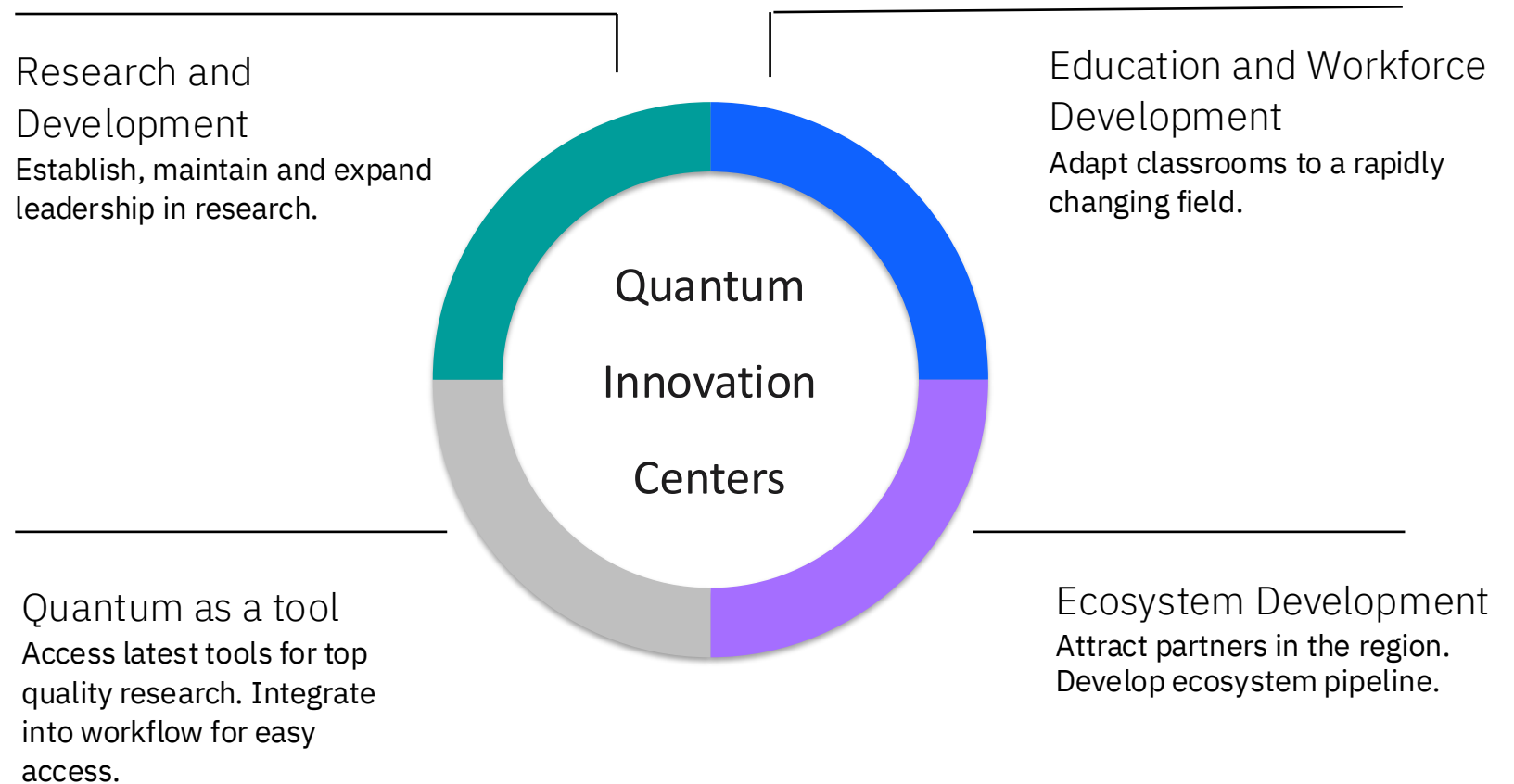


IBM Quantum
Advanced



Quantum Innovation Centers (QIC)

*Strategic partners developing regional and national leadership in quantum computing by leveraging IBM Quantum capabilities and expertise. QICs act as **Ecosystem Builders**, bringing together academia, industry, and government.*



Road to Practitioner Program in Canada

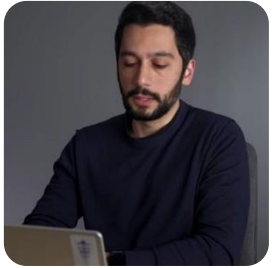
Partner (QIC): PINQ

IBM Engagement Lead: Camille Brun-Jolicouer



Road to Practitioner Team

Main lecturers:



Ibrahim Shehzad



Sean Wagner

Technical support:



Sophie Engineer



Jorge Martinez



Marcel Pfaffhuser

Communication and coordination:



Camille Brun-Jolicouer

Program coordination



Fabio Scafirimuto

Introduction to Quantum Computing

The **new** wave of computing



Classical computer

Well suited for many problems



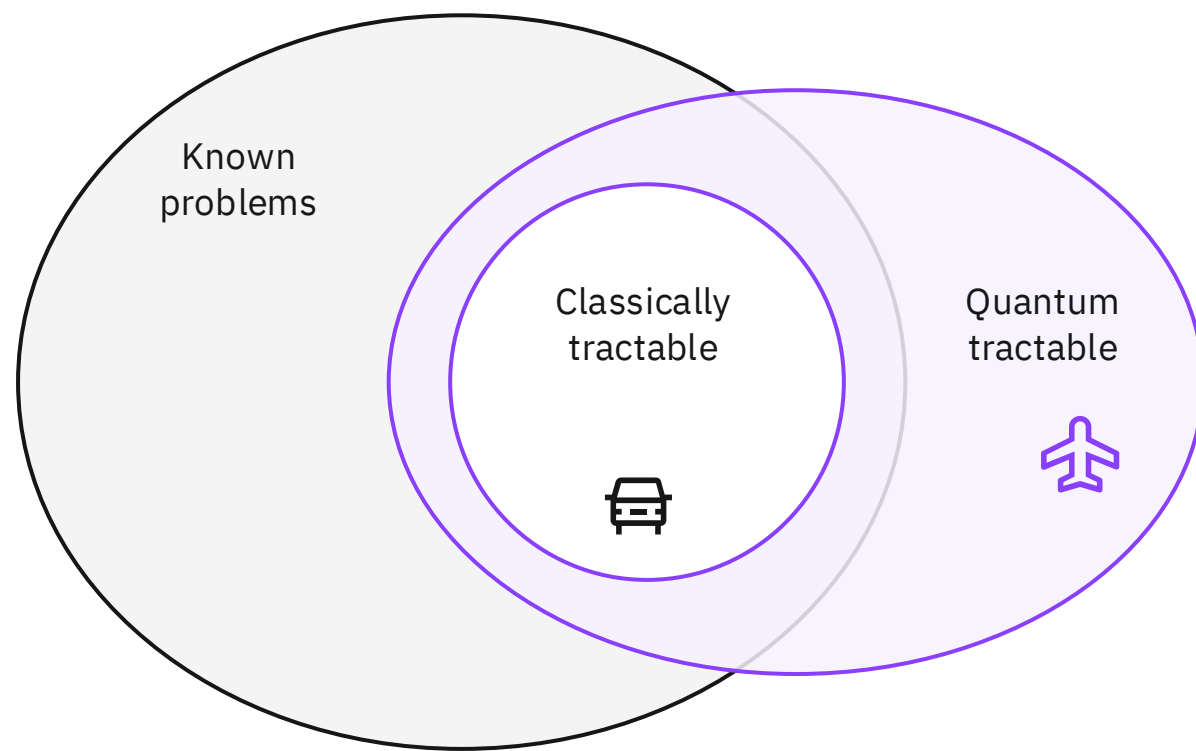
Quantum computer

Unlock classically intractable problems

Unlock discovery with quantum computing

Harnesses the capacity to advance conceptual and tools-based discovery

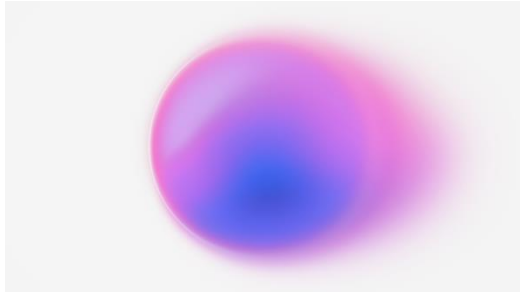
Quantum computers are **exponentially more powerful** than classical computers



1. **A new way of computing**
New paradigm of problem-solving and thinking
2. **Solving new problems**
Unlock classically unsolvable problems, cutting computation time down from hours to minutes
3. **Discovery of new use cases**
Expand discovery into new computational spaces

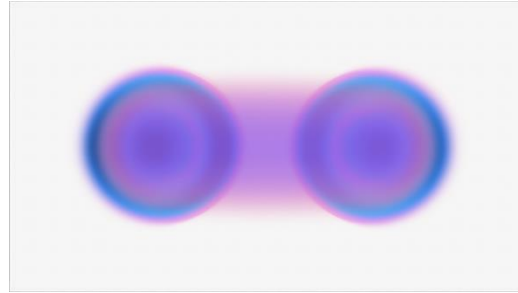
Uniquely quantum

Some problems are classically intractable and will never be solvable with traditional computers



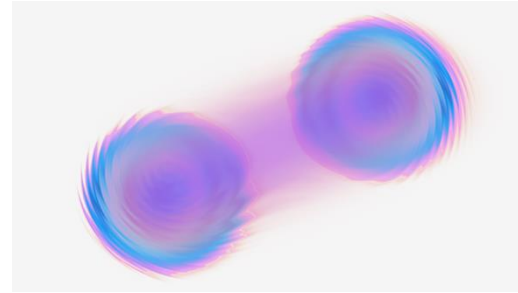
Superposition

A quantum system existing in a complex linear combination of two states until it is measured



Entanglement

Information shared jointly between entangled pairs or groups



Interference

Interaction that affects likelihood of solutions

Moore's law: the number of transistors in a classical integrated circuit doubles about every two years ... but we are approaching the end due to physical limitations

[Approaching the physical limit: IBM created the world's first 2 nm node chip in 2021, with transistors as small as 10 silicon atoms](#)

These Quantum concepts can reduce the number of computational steps required for certain algorithms &

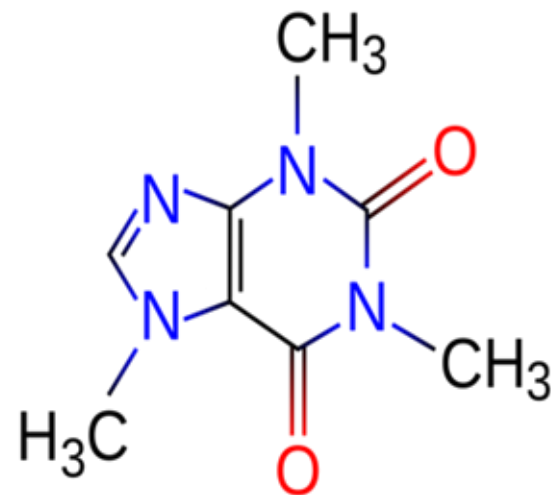
At a ~100 qubit scale, with sufficient circuit depth and complexity, classical computers can no longer simulate exactly

Computing with caffeine

If our best classical computers are so powerful, shouldn't we be able to perfectly simulate molecules and chemical reactions?

This would allow us to accelerate discovery of new compounds and processes for healthcare, materials, alloys, and sustainable energy creation.

Let's consider caffeine ...



Computing with caffeine

We would need approximately 10^{48} bits to represent the energy configuration of a single caffeine molecule at a single instant in a classical computer.

This is 1 to 10% of the total number of atoms in the Earth.

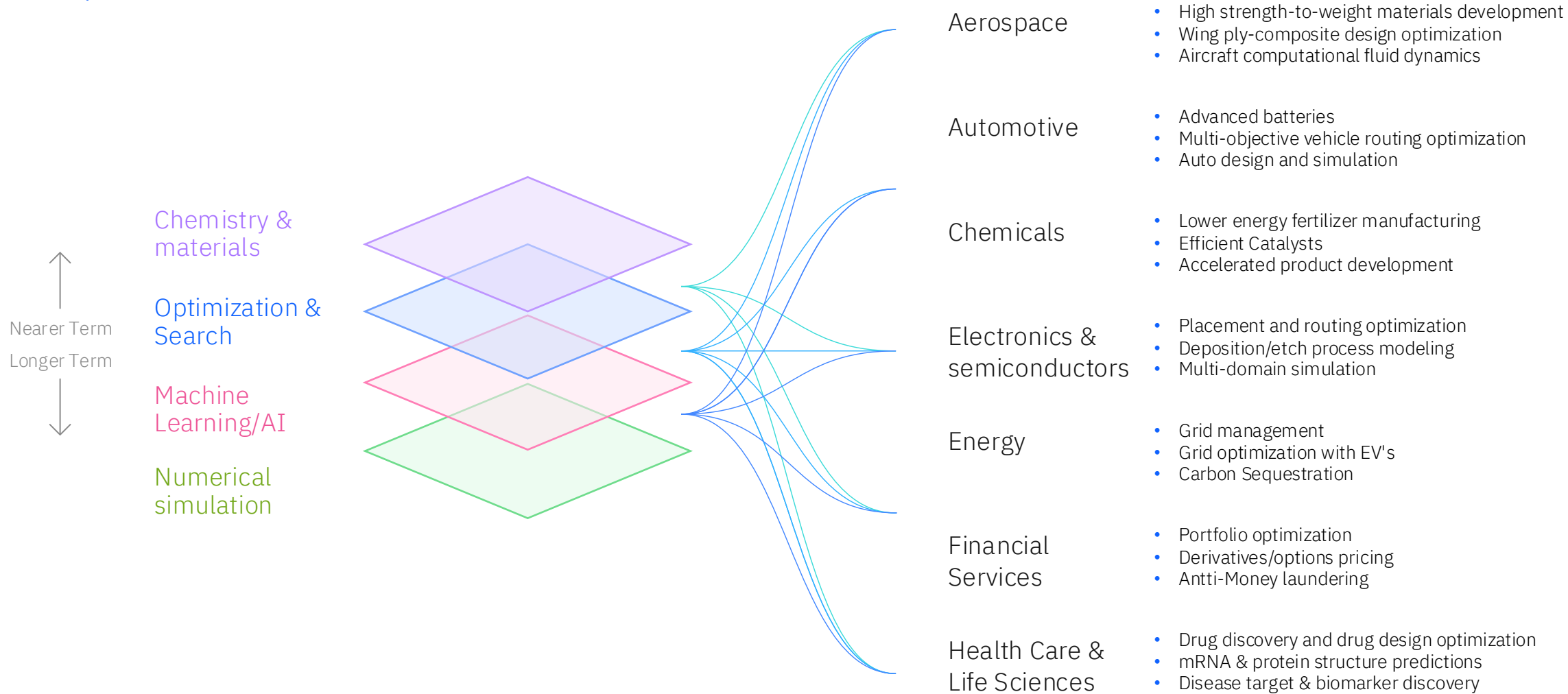
$10^{48} =$ 1,000,000,000,000,000,
000,000,000,000,000,000,
000,000,000,000,000

Although it's impossible to completely represent the molecular configuration of caffeine on today's most powerful super computers, we could represent it using 160 error-corrected qubits.

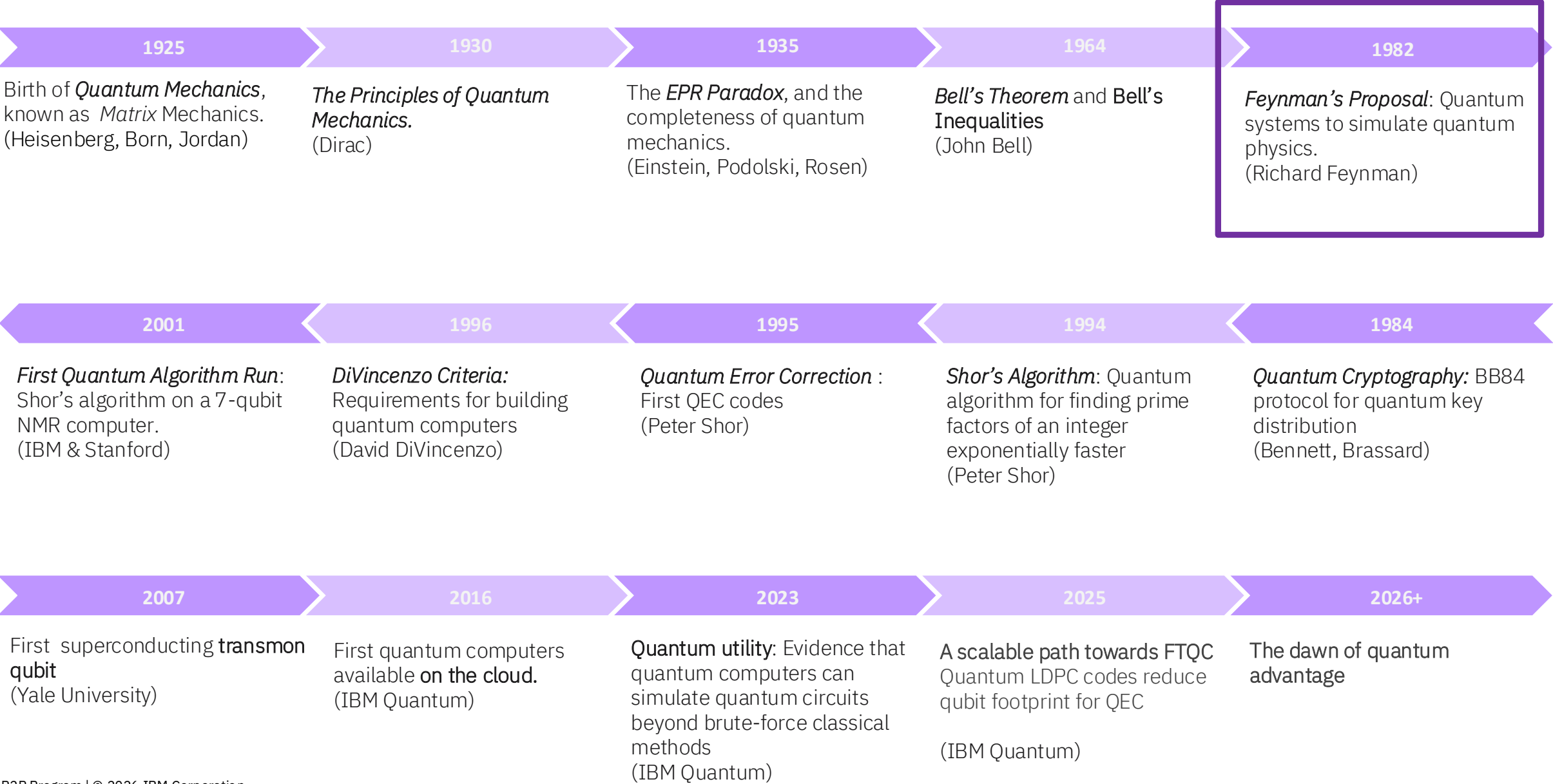


Quantum computing is expected to have impact across industries

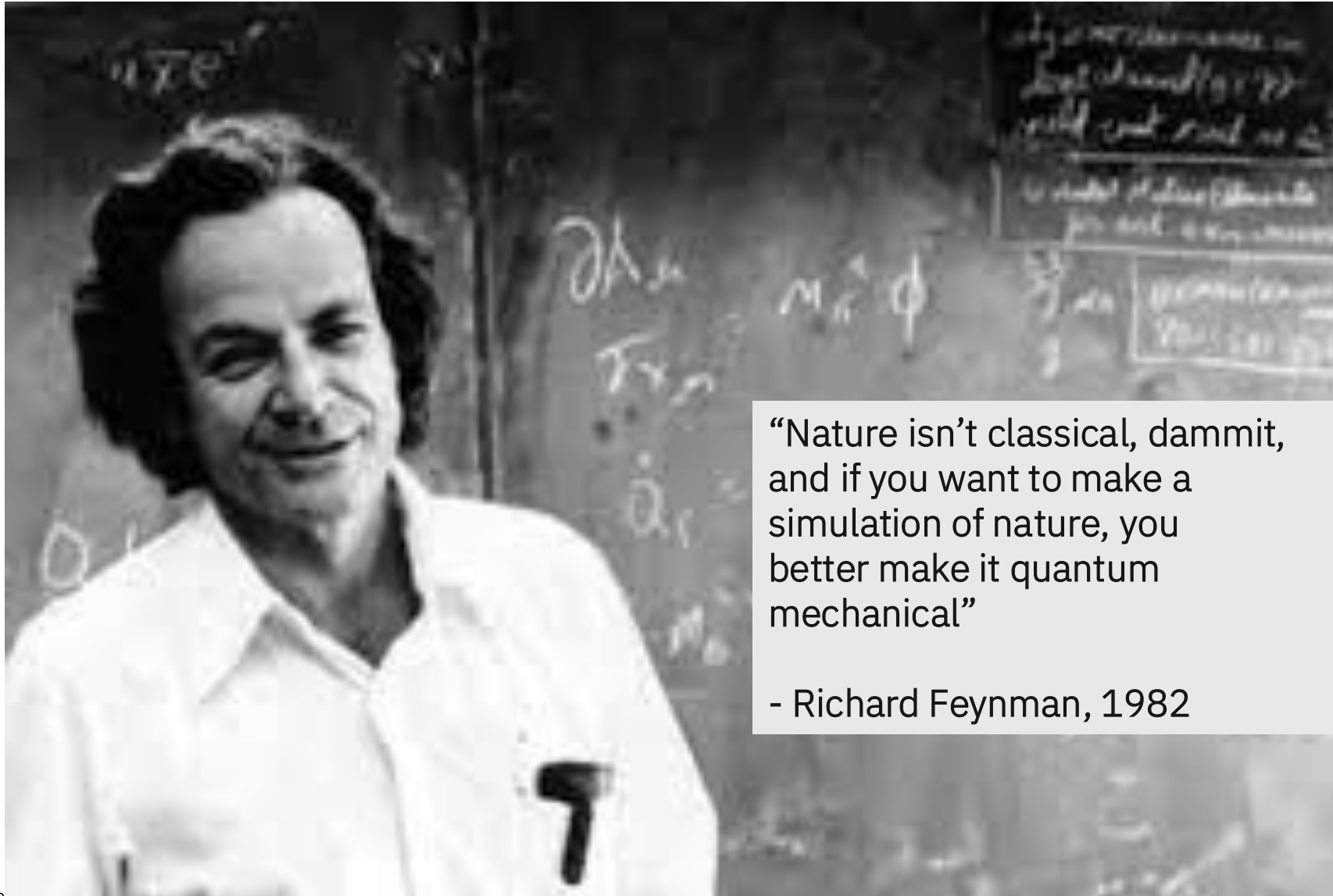
Example use cases



Some Milestones in the History of Quantum Computing



A change of paradigm



“Nature isn’t classical, dammit,
and if you want to make a
simulation of nature, you
better make it quantum
mechanical”

- Richard Feynman, 1982

1982

1984

2026+

Quantum Computing Basics

Bits and classical logic circuits

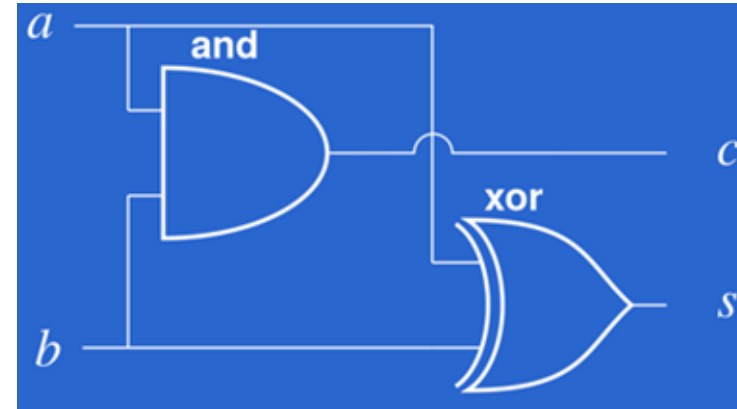
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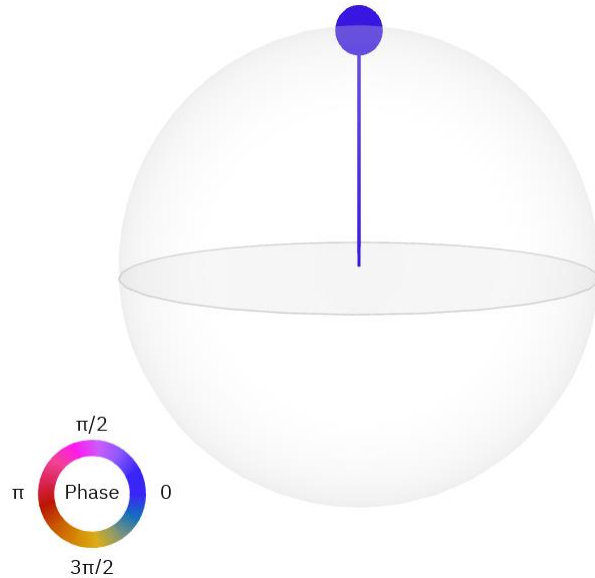
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A **bit** is a controllable classical object that is the unit of information

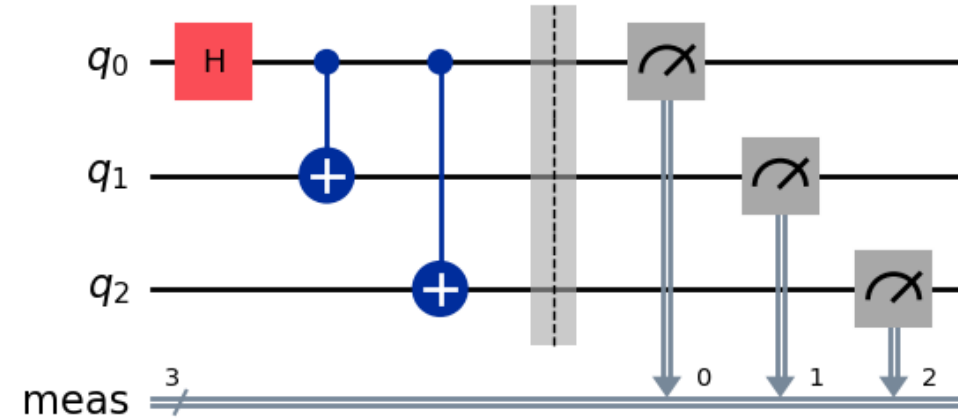


A **classical logic** circuit is a set of gate operations on bits and is the unit of computation

Quantum bits (qubits) and quantum circuits



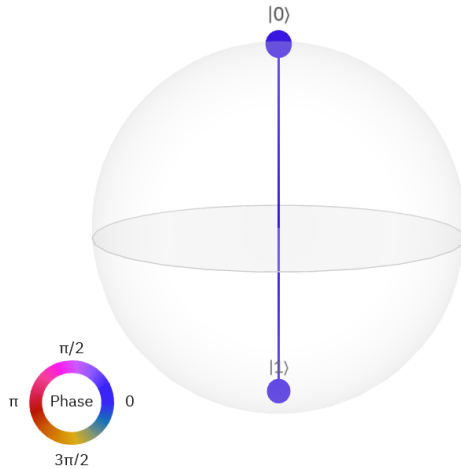
A **quantum bit** or qubit is a controllable quantum object that is the unit of information



A **quantum** circuit is a set of quantum gate operations on qubits and is the unit of computation

Quantum computing uses essential ideas from quantum mechanics

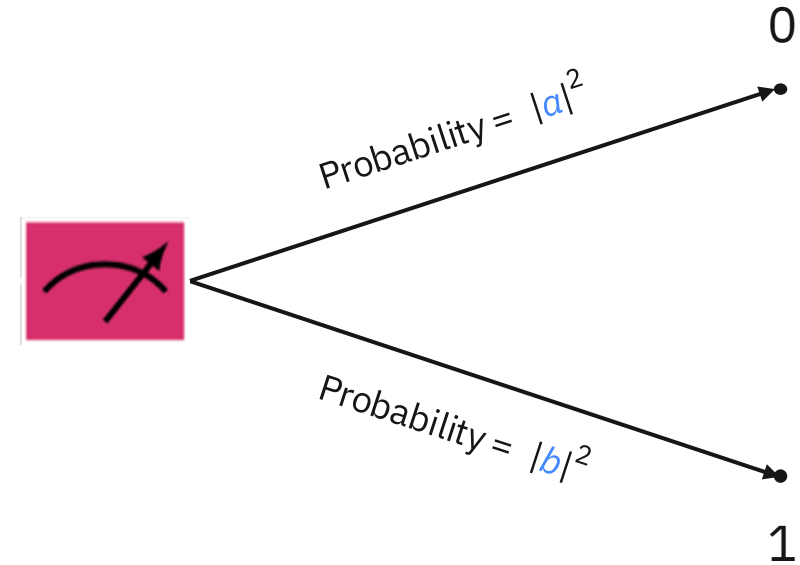
Measurements



A qubit's **state** is a combination of $|0\rangle$ and $|1\rangle$:

$$a |0\rangle + b |1\rangle$$

This means that a single qubit contains
two pieces of information.



When we measure a qubit, the result comes
with a certain probability:

$|a|^2$ is the probability we will get $|0\rangle$

$|b|^2$ is the probability we will get $|1\rangle$

Quantum computing
uses essential ideas from quantum mechanics

Superposition

With one qubit

$$a |0\rangle + b |1\rangle$$

$$|a|^2 + |b|^2 = 1$$

With two qubits we get combinations like

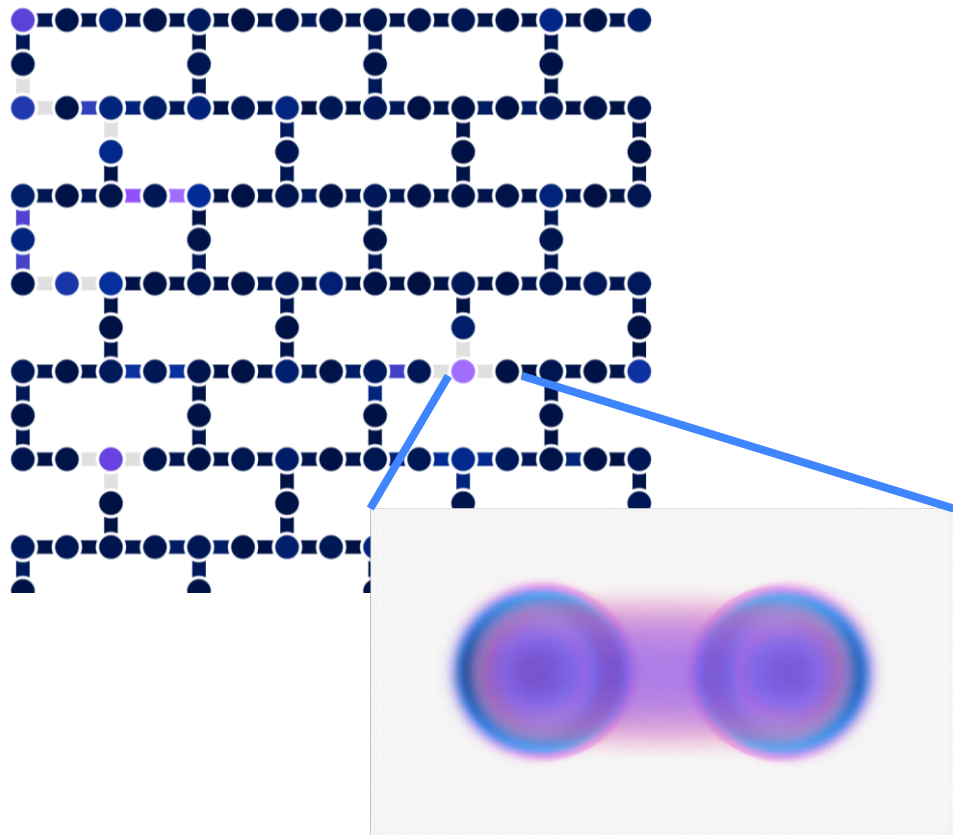
$$a |00\rangle + b |01\rangle + c |10\rangle + d |11\rangle$$

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

(where $|01\rangle$ means the first qubit is $|0\rangle$ and the second is $|1\rangle$).

Quantum computing
uses essential ideas from quantum mechanics

Entanglement



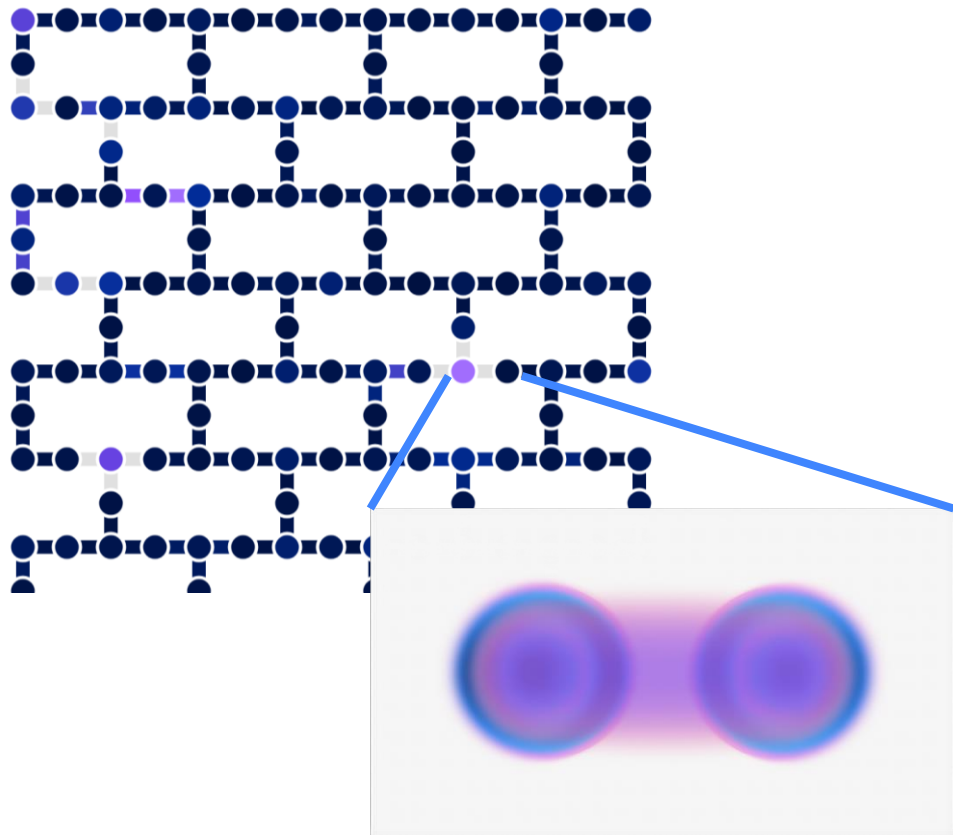
When the state of two or more qubits is not equivalent to the product of his components, we say that the qubits are **entangled**

$$a |00\rangle + b |01\rangle + c |10\rangle + d |11\rangle$$

If two or more of the a , b , c , and d are non-zero, and we cannot separate the qubits, then they are **entangled** with perfect correlation and are no longer independent.

Quantum computing uses essential ideas from quantum mechanics

Entanglement



Examples:

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle \quad \text{not entangled}$$

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |11\rangle \quad \text{entangled}$$

In fact, we can write

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle = |0\rangle \left(\frac{\sqrt{2}}{2} |0\rangle + \frac{\sqrt{2}}{2} |1\rangle \right)$$

but we cannot write

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |11\rangle$$

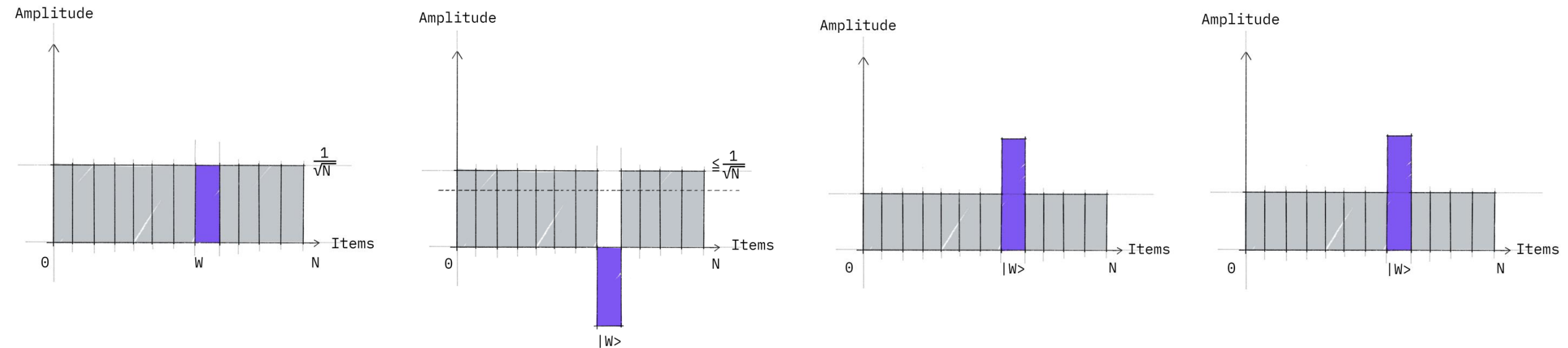
as the “product” of two single qubit states. Therefore, the qubits are **entangled**!

Once you measure the first qubit, the second is uniquely determined.

Quantum computing uses essential ideas from quantum mechanics

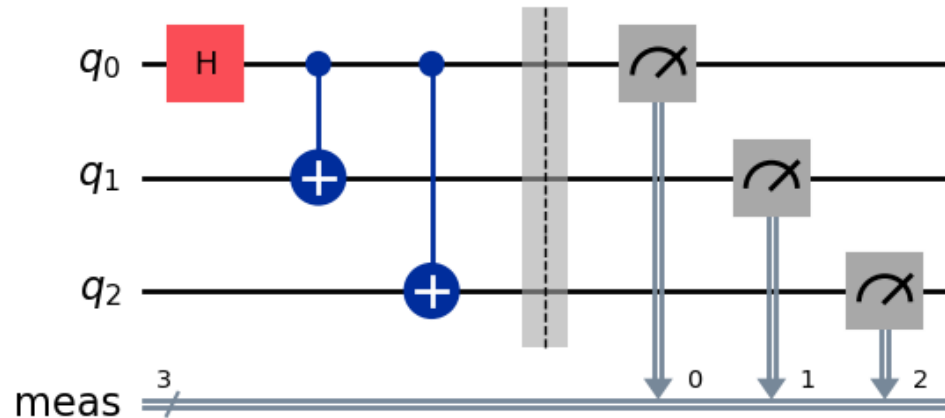
Interference

Interference allows us to increase the probability of getting the right answer and decrease the chance of getting the wrong one.



Gates: minimum unit of computation

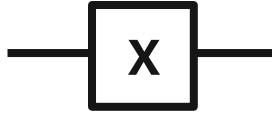
Gates / operations



Classical logical circuits use operations like [and](#), [or](#), [not](#), [nand](#), and [xor](#). We also call these gates.

Quantum circuits use reversible gates that change the quantum states of one, two, or more qubits.

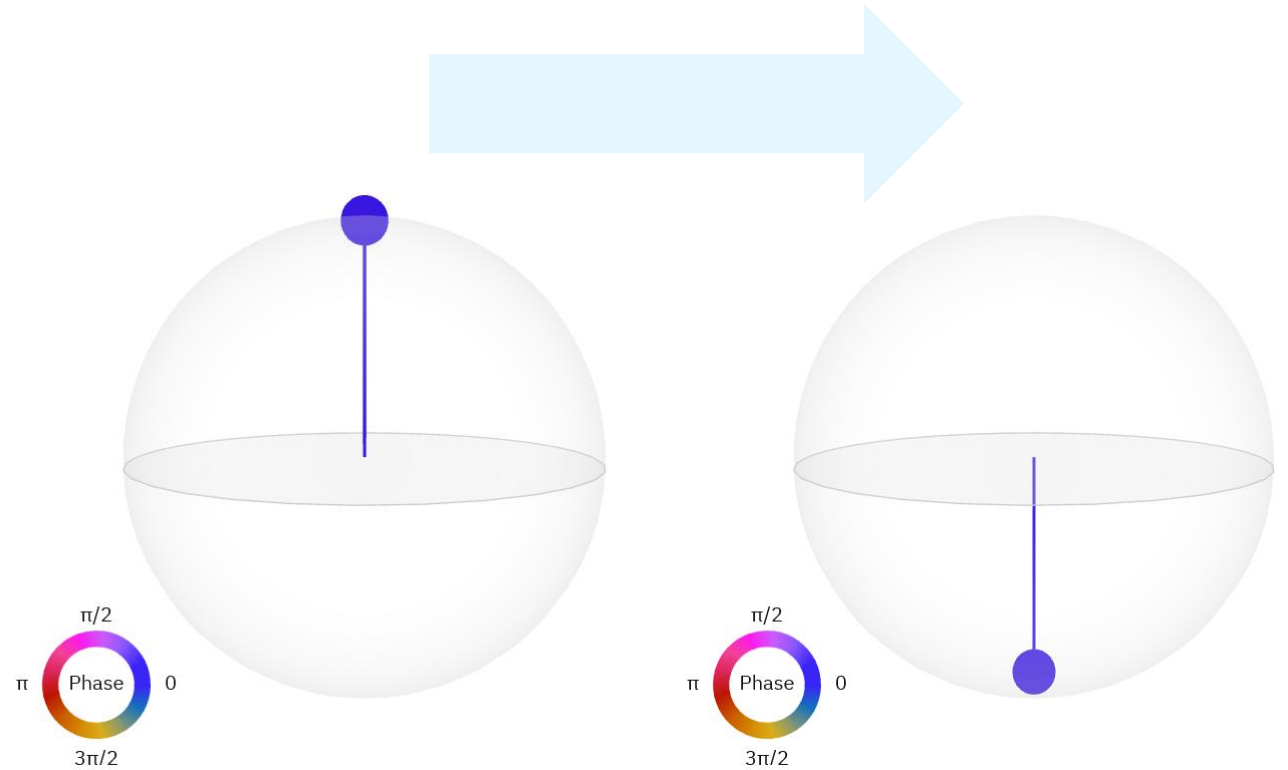
Gates: the effect of the X gate



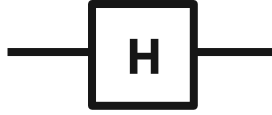
The Pauli **X** gate, also known as the NOT gate, flips $|0\rangle$ into $|1\rangle$ and vice versa:

$$a |0\rangle + b |1\rangle \mapsto b |0\rangle + a |1\rangle$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}$$



Gates: the effect of the Hadamard gate



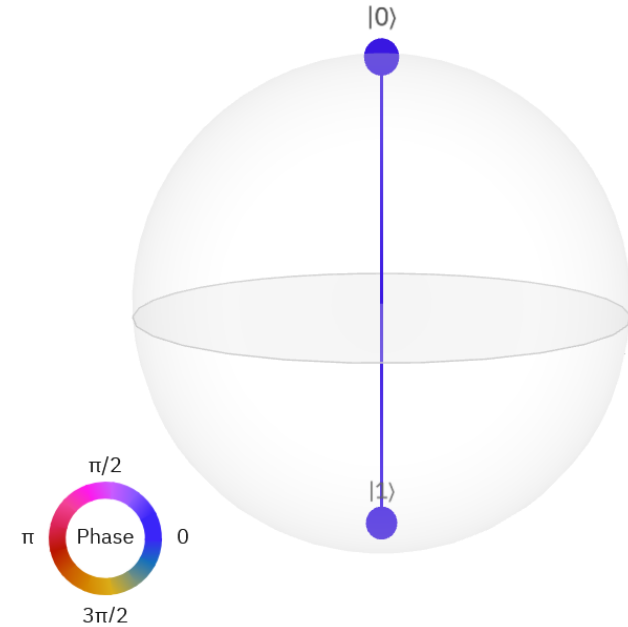
The **H** gate creates superposition:

$$a |0\rangle + b |1\rangle$$

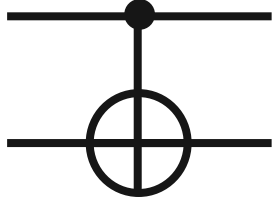
$a = 1$ and $b = 0$, so $|0\rangle$ is mapped to

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle).$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a + b \\ a - b \end{pmatrix}$$



Gates: the effect of the **CNOT** gate



The **CNOT** gate makes a controlled operation.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} a \\ b \\ d \\ c \end{pmatrix}$$

$$|\psi\rangle = |00\rangle$$

$$|\psi\rangle = |01\rangle$$

$$|\psi\rangle = |10\rangle$$

$$|\psi\rangle = |11\rangle$$



$$|\psi\rangle = |00\rangle$$

$$|\psi\rangle = |01\rangle$$

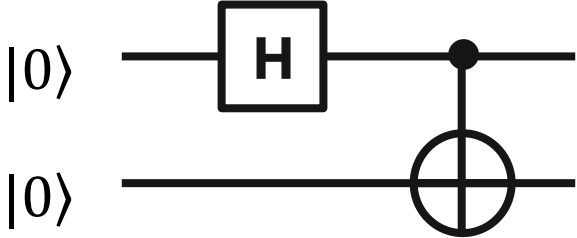
$$|\psi\rangle = |11\rangle$$

$$|\psi\rangle = |10\rangle$$

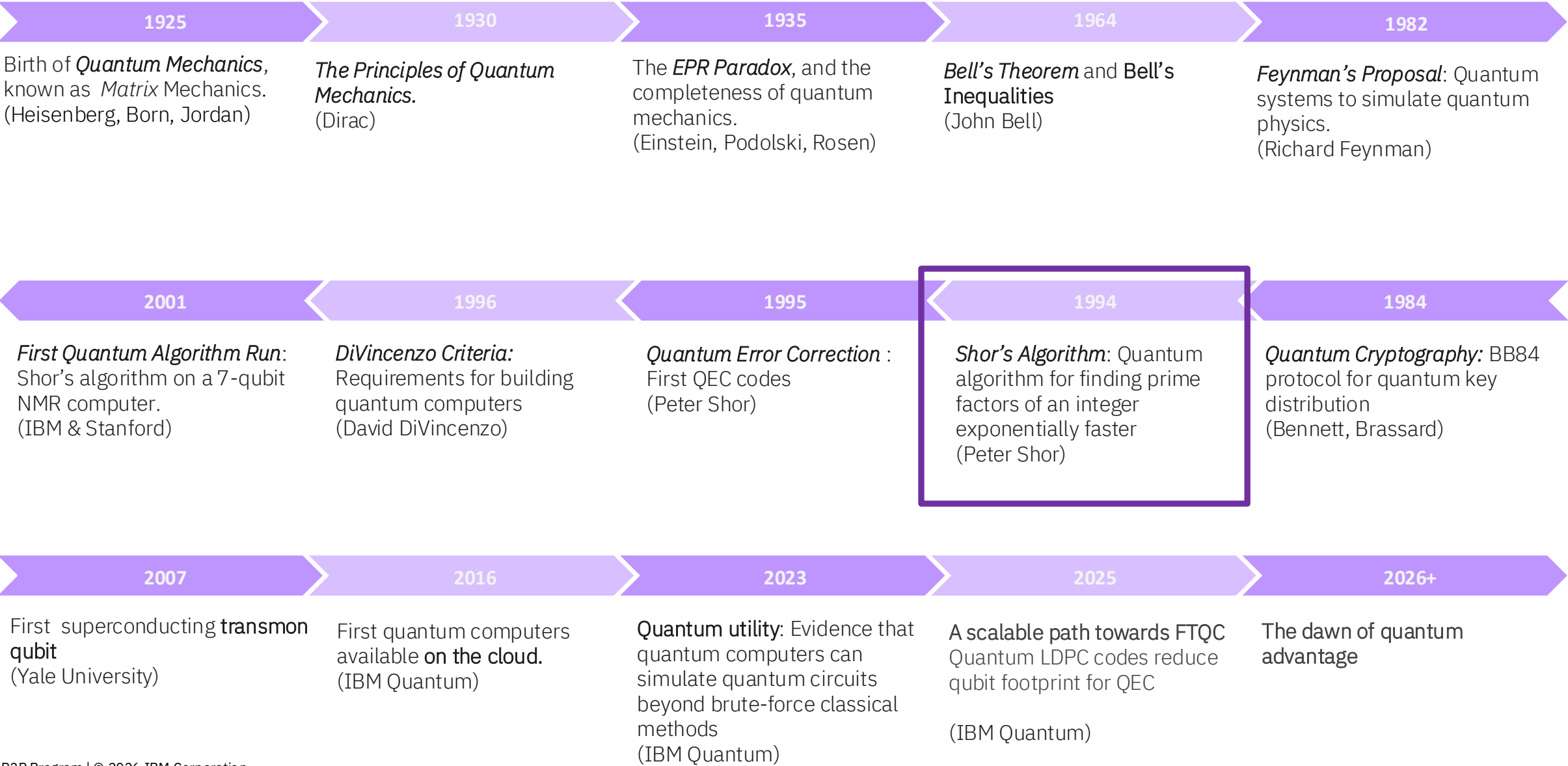
Gates: Creating a Bell State

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Notice that the application of Hadamard and CNOT gates create entangled states

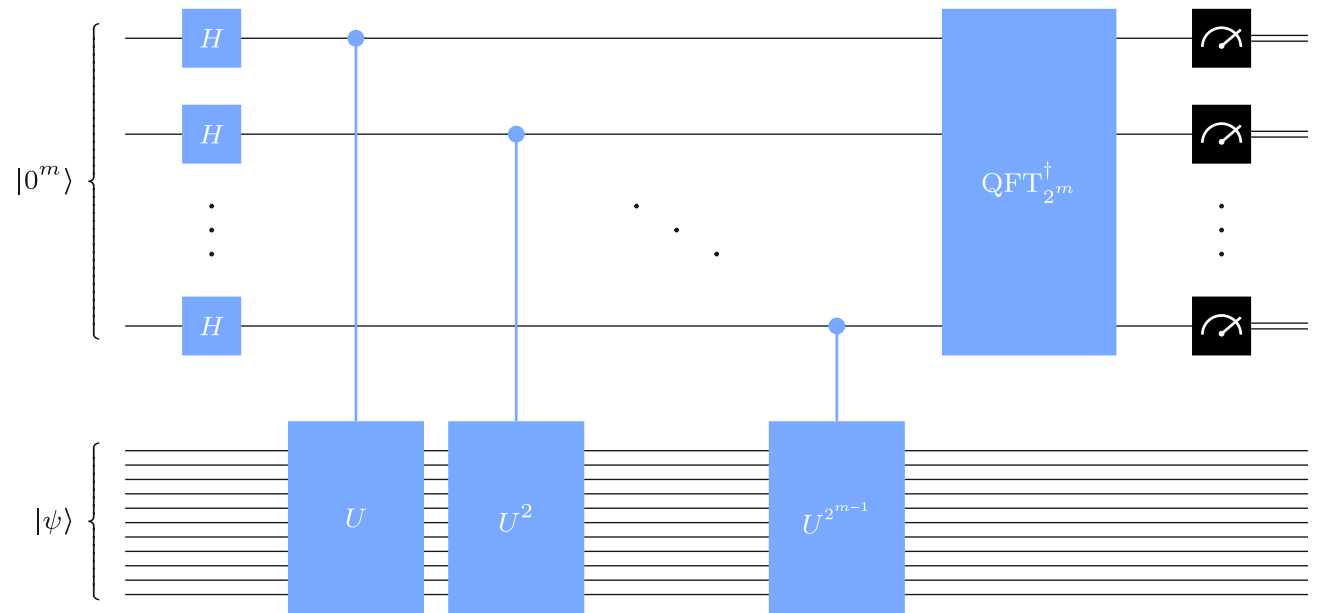

$$\left. \begin{array}{l} |0\rangle \\ |0\rangle \end{array} \right\} (H|0\rangle)|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$$
$$CNOT \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Some Milestones in the History of Quantum Computing

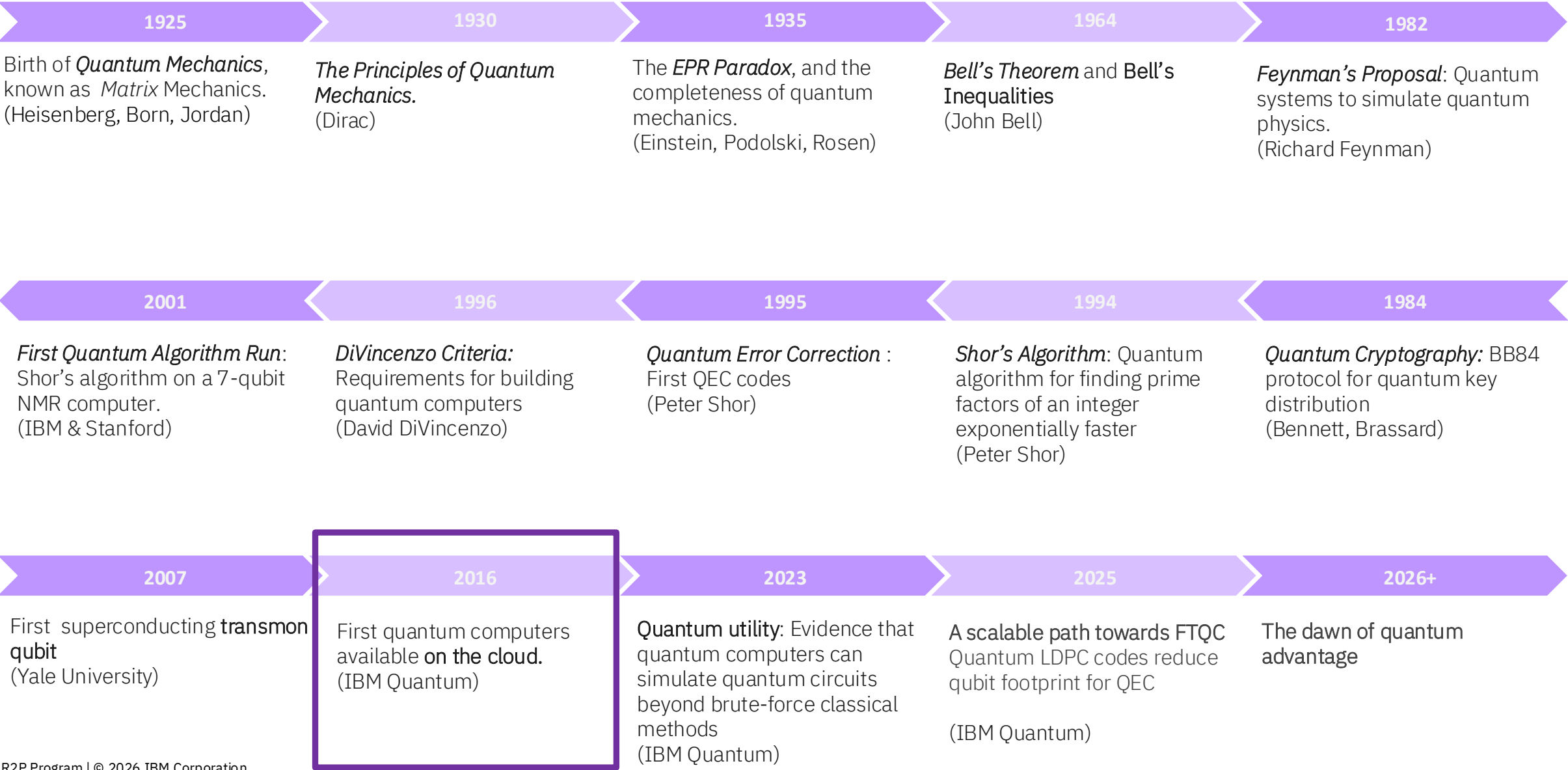


Shor's algorithm

- A quantum algorithm to factor large integers into prime numbers
- Exponentially faster than any other classical algorithm
- Revolution in cryptography such as RSA that relies on prime factorization
- Reduces integer factorization into finding the period of a function $f(x) = a^x \bmod N$ using Quantum Phase Estimation
- Needs ~ 1M physical qubits to quantum advantage



Some Milestones in the History of Quantum Computing



IBM Just Put A Quantum Computer On The Cloud For Anyone To Use

By [Alex Konrad](#), Former Staff. Senior editor covering venture capital and startups

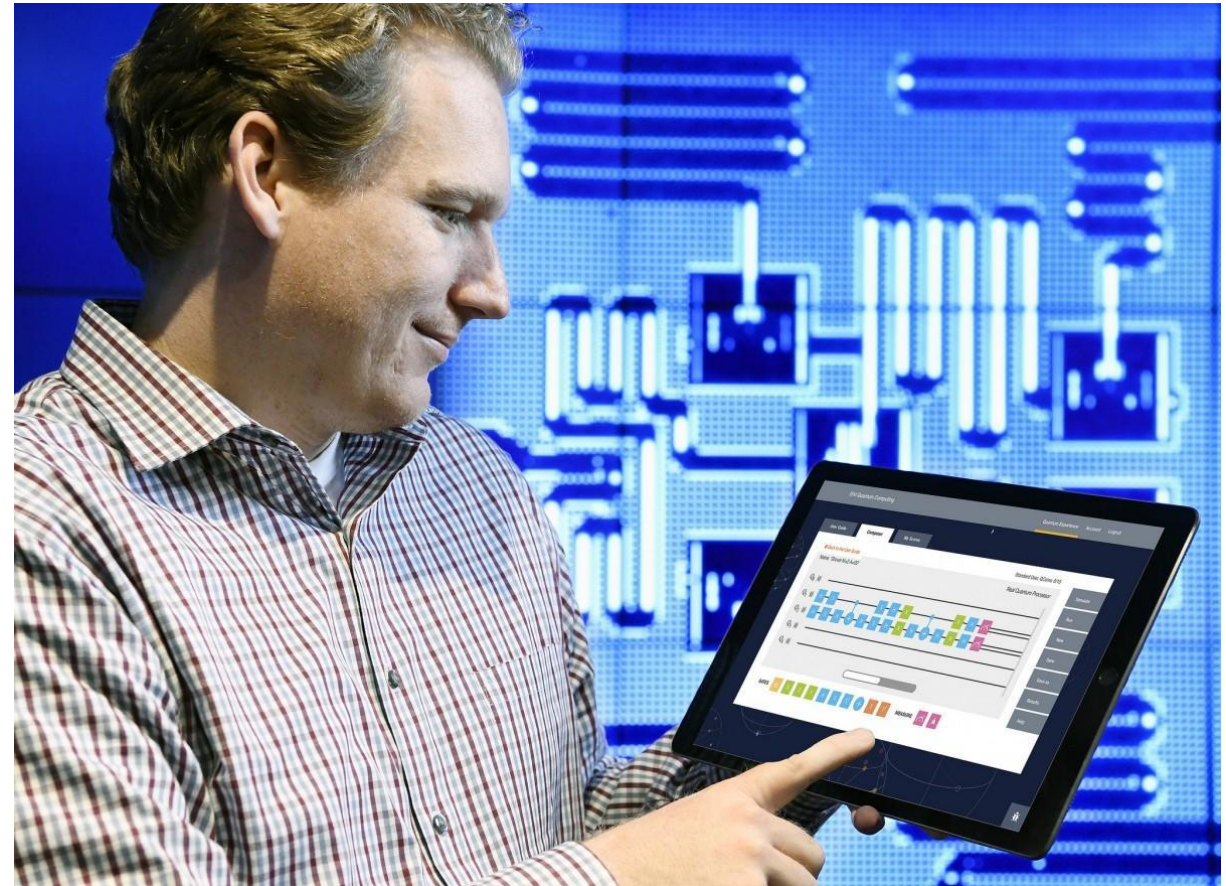
Published May 04, 2016, 12:14am EDT, Updated May 04, 2016, 08:07pm EDT



IBM's New Quantum Computing Move Marks A New Era For Cloud Computing

By [Greg Satell](#), Former Contributor.

Published May 04, 2016, 12:03am EDT, Updated May 18, 2016, 07:47am EDT



Since 2016 IBM has
publicly deployed:

60

quantum
devices

(<100 qubits)

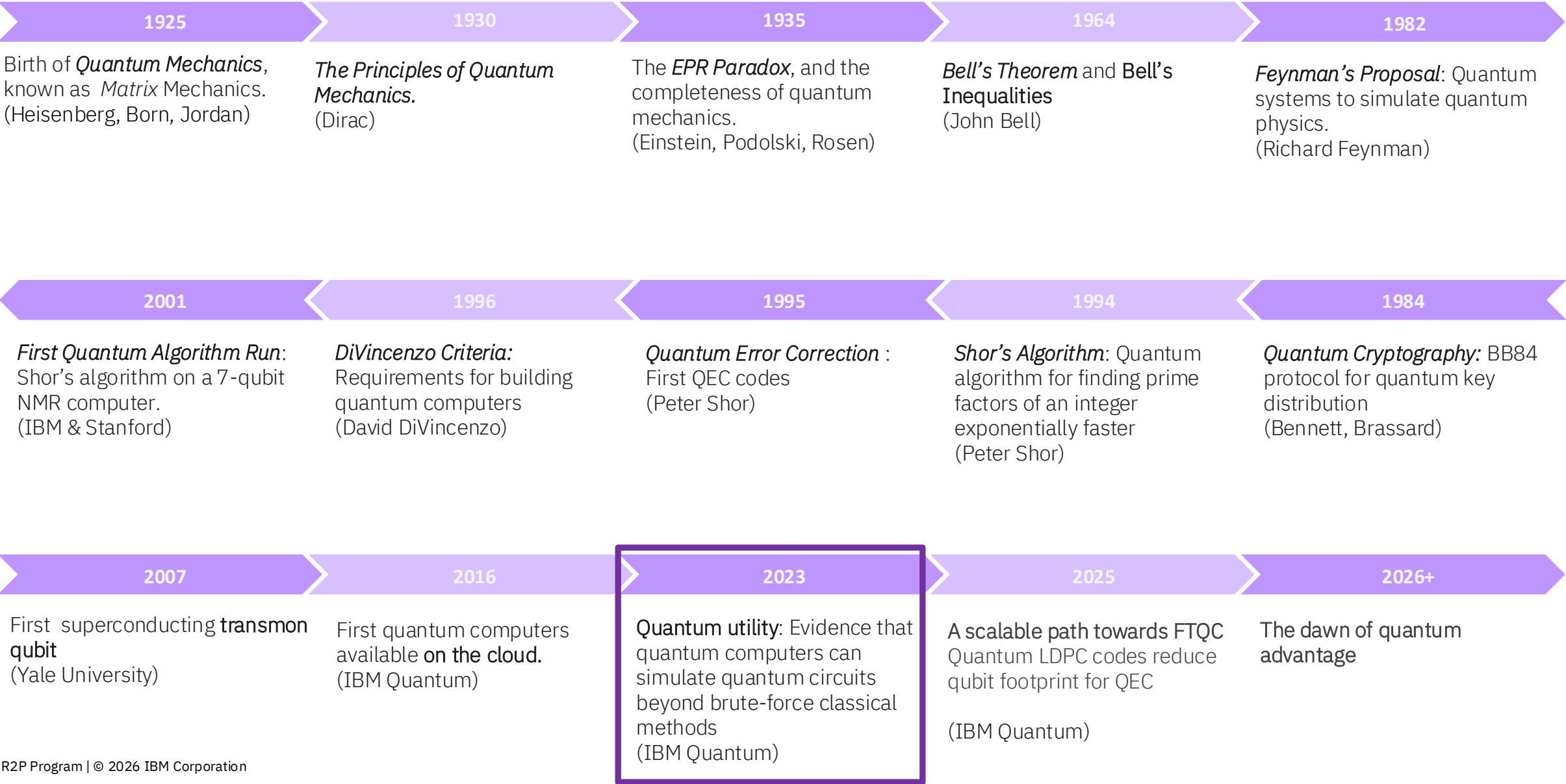
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quantum computers

(>100 qubits) that can run
circuits with >5000 two-qubit gates

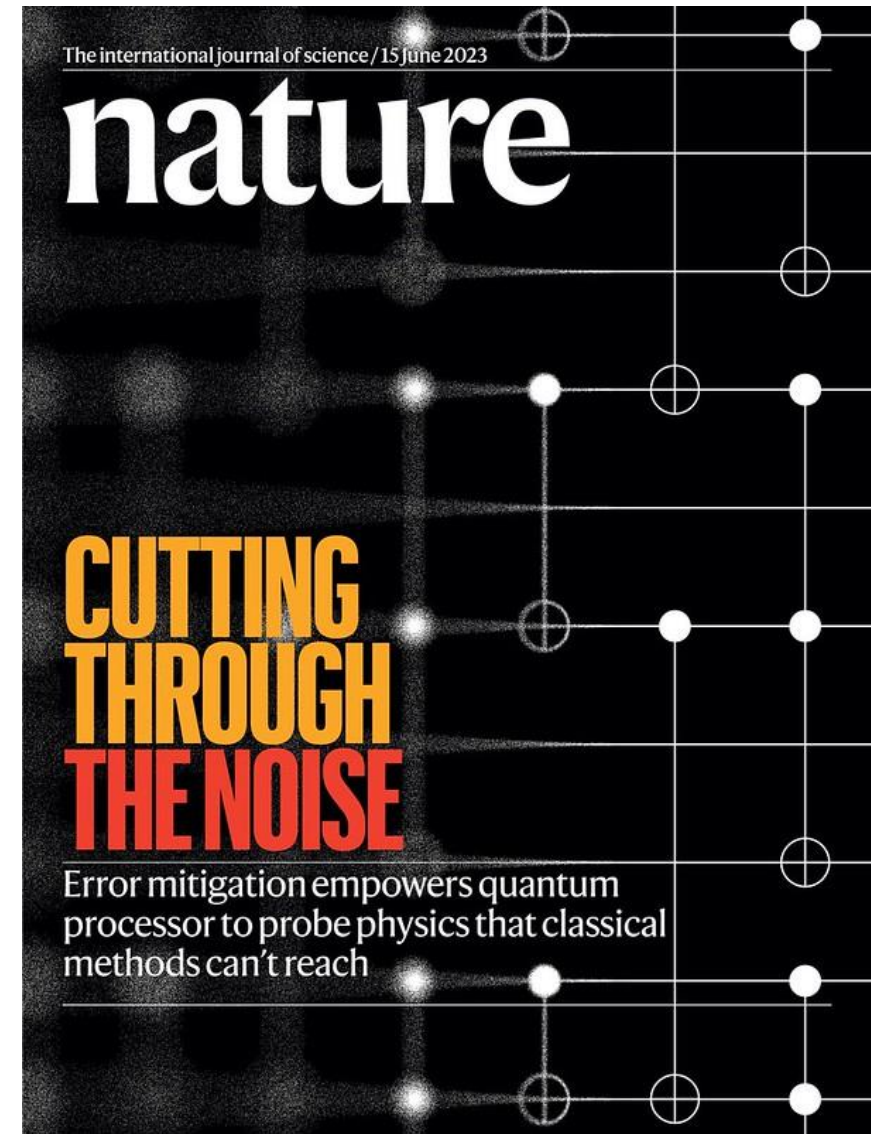


Some Milestones in the History of Quantum Computing

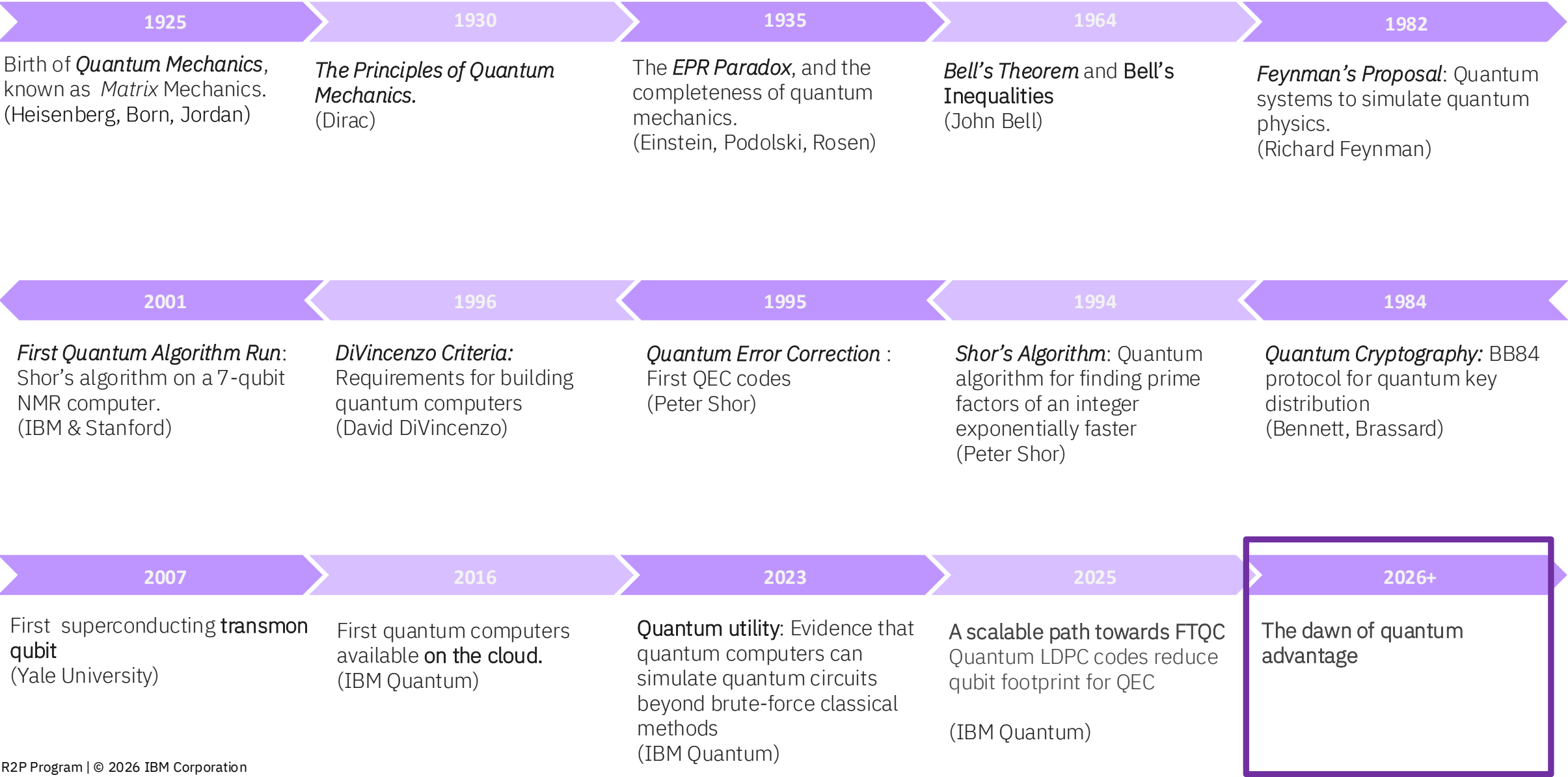


Evidence of utility before fault tolerance

- IBM Quantum and UC Berkeley demonstrated that quantum is competitive with leading classical methods.
- Demonstrated IBM Quantum computers can produce accurate expectation values beyond the reach of brute-force classical computation.
- IBM Quantum has the trinity of **scale, quality and speed** needed for demonstrations of useful quantum computation.
- The IBM Quantum mission is to bring useful quantum computing to the world.

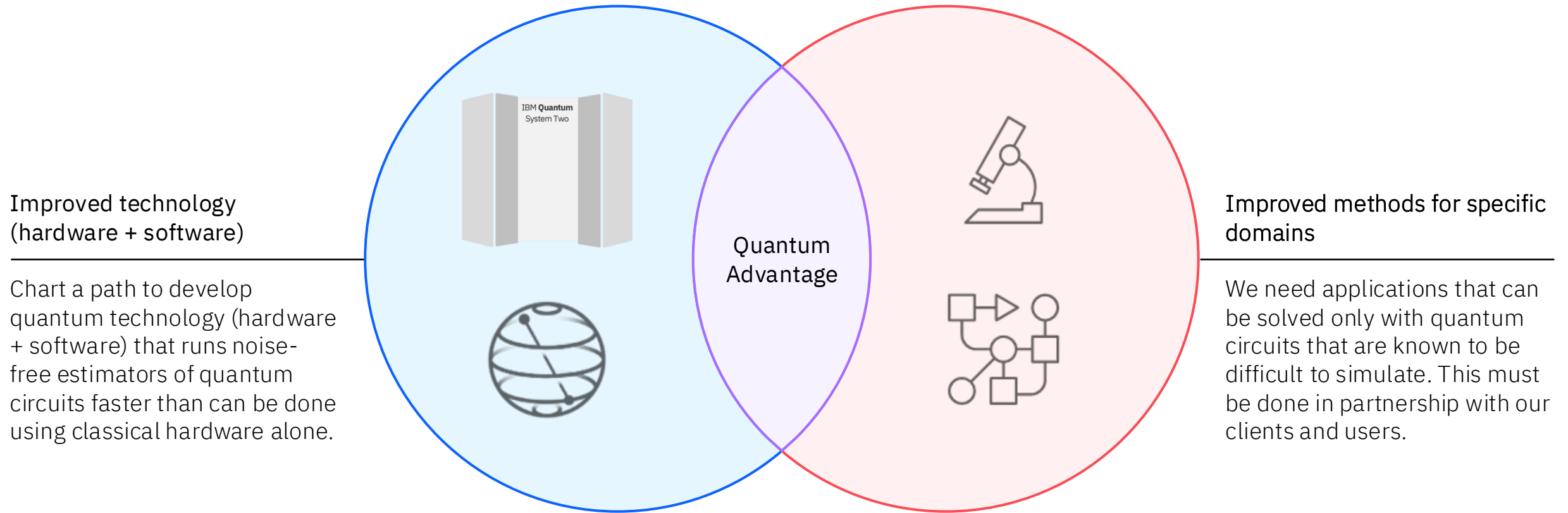


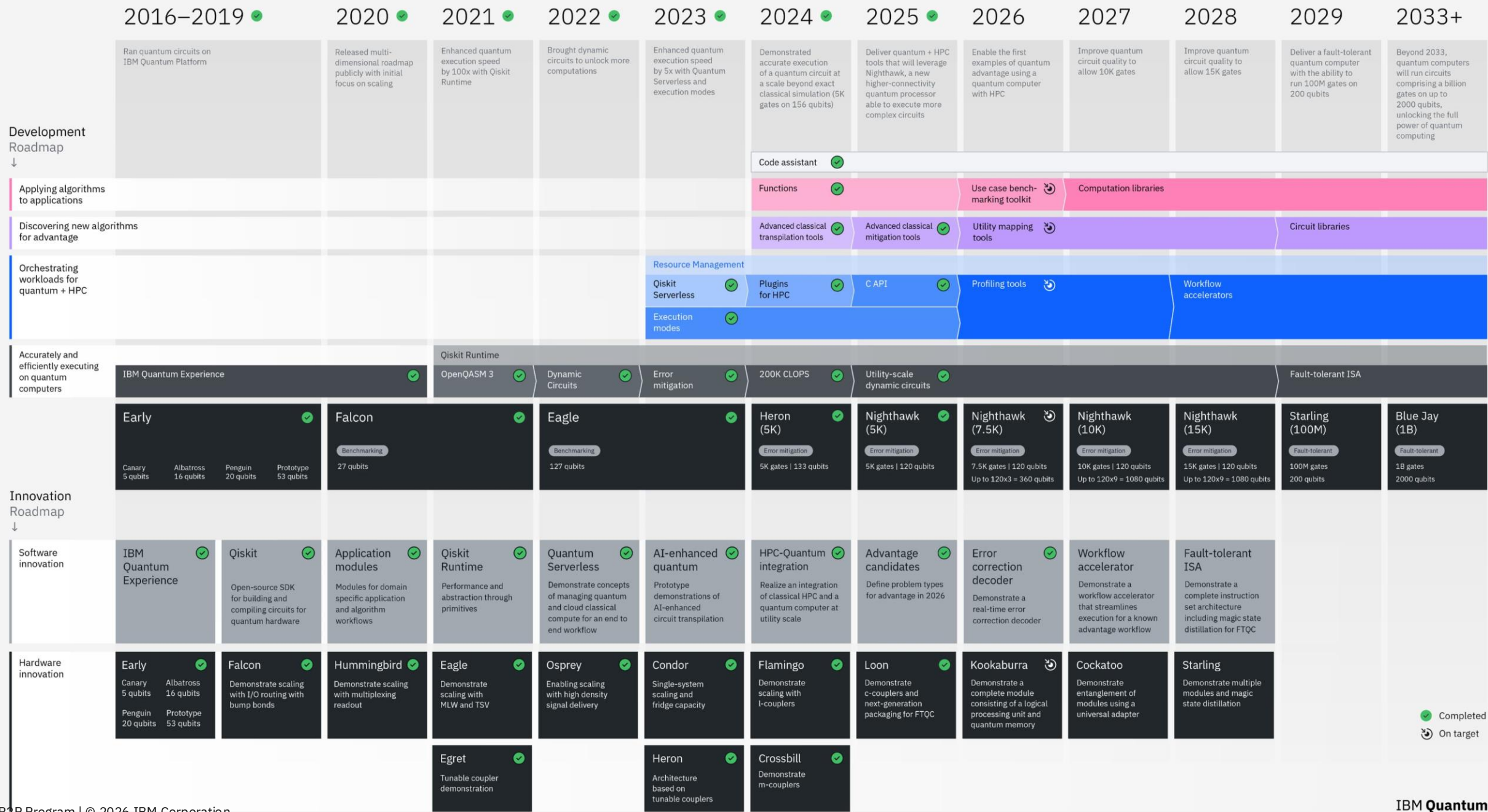
Some Milestones in the History of Quantum Computing



Towards Quantum Advantage

The path to quantum advantage





Completed
On target

Key milestones in bringing useful quantum computing to the world

2023

Establish quantum utility



2026

Demonstrate quantum advantage



2029

Deliver the first large-scale, fault-tolerant quantum computer



Key milestones in bringing useful quantum computing to the world

2023

Establish quantum utility



2026

Demonstrate quantum advantage



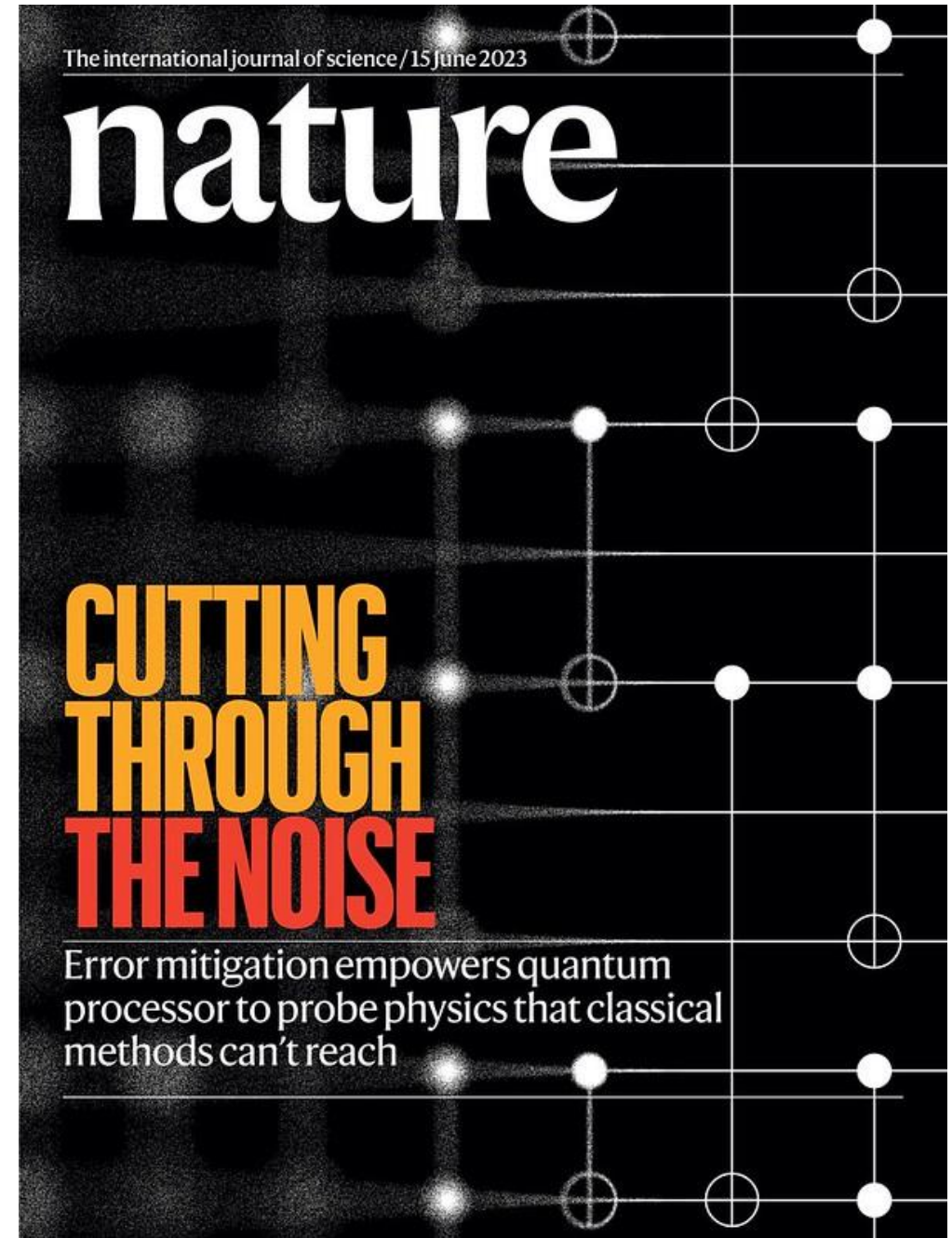
2029

Deliver the first large-scale, fault-tolerant quantum computer

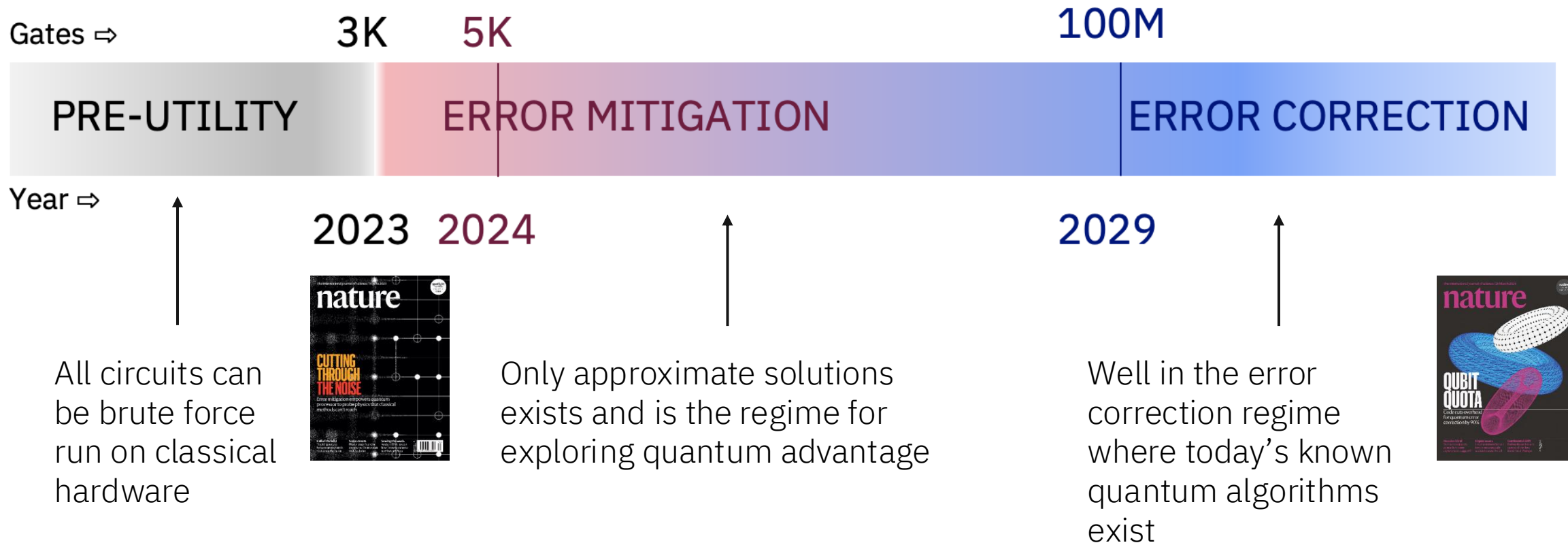


A noisy quantum
computer produces
accurate
expectation values
on 127 qubits and 2880
gates, outside of brute
force classical
computation.

Y. Kim, A. Eddins, et al, Nature. 618, 500–505 (2023)



Quantum Utility

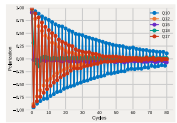


If you build it, they will come... (1/2)



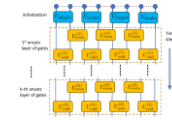
Evidence for the utility of quantum computing before fault tolerance
Nature, **618**, 500 (2023)
127 qubits / 2880 CX gates

simulation



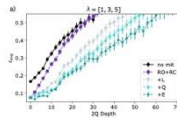
Characterizing quantum processors using discrete time crystals
arXiv:2301.07625
80 qubits / 7900 CX gates

simulation



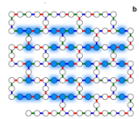
Simulating large-size quantum spin chains on cloud-based superconducting quantum computers
Phys. Rev. Research **5**, 013183 (2023)
102 qubits / 3186 CX gates

simulation



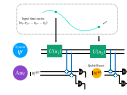
Best practices for quantum error mitigation with digital zero-noise extrapolation
arXiv:2307.05203
104 qubits / 3605 ECR gates

tools



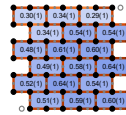
Uncovering Local Integrability in Quantum Many-Body Dynamics
arXiv:2307.07552
124 qubits / 2641 CX gates

simulation



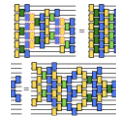
Quantum reservoir computing with repeated Measurements on superconducting devices
arXiv:2310.06706
120 qubits / 49470 ECR gates + meas.

QML



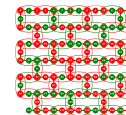
Realizing the Nishimori transition across the error threshold for constant-depth quantum circuits
arXiv:2309.02863
125 qubits / 429 gates + meas.

simulation



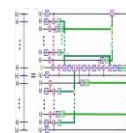
Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits
PRX Quantum **5**, 020315 (2024)
100 qubits / 788 CX gates

simulation



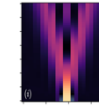
Scaling Whole-Chip QAOA for Higher-Order Ising Spin Glass Models on Heavy-Hex Graphs
arXiv:2312.00997
127 qubits / 420 CX gates

optimization



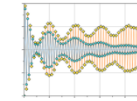
Efficient Long-Range Entanglement using Dynamic Circuits
PRX Quantum **5**, 030339 (2024)
101 qubits / 504 ECR gates + meas

tools



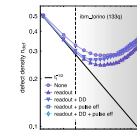
Quantum Simulations of Hadron Dynamics in the Schwinger Model using 112 Qubits
Phys. Rev. D **109**, 114510 (2024)
112 qubits / 13,858 CZ gates

simulation



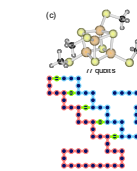
Unveiling clean two-dimensional discrete time quasicrystals on a digital quantum computer
arXiv:2403.16718
133 qubits / 15,000 CZ gates

simulation



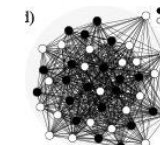
Benchmarking digital quantum simulations and optimization above hundreds of qubits using quantum critical dynamics
arXiv:2404.08053
133 qubits / 1440 CX gates

simulation



Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer
arXiv:2405.05068
77 qubits / 3590 CZ gates

simulation



Towards a universal QAOA protocol: Evidence of quantum advantage in solving combinatorial optimization problems
arXiv:2405.09169
109 qubits / 21,200 ECR gates

optimization

Key milestones in bringing useful quantum computing to the world

2023

Establish quantum utility



2026

Demonstrate quantum advantage



2029

Deliver the first large-scale, fault-tolerant quantum computer



A framework for quantum advantage

<https://arxiv.org/abs/2506.20658>

Several experimental demonstrations have shown that quantum processors can now perform accurate computations at scales that **surpass the capabilities of brute-force classical simulation** of quantum circuits.

We propose a **functional definition of quantum advantage** and delineate the criteria required to substantiate such a claim.

Establishing such criteria for when and how quantum advantage is achieved is crucial, not only as a rigorous benchmark of quantum technological progress, but also for **identifying application domains with the potential for near-term impact**.

A framework for quantum advantage

O. Lanes,¹ M. Beji,² A. D. Córcoles,¹ C. Dalyac,² J. M. Gambetta,¹ L. Henriet,² A. Javadi-Abhari,¹ A. Kandala,¹ A. Mezzacapo,¹ C. Porter,¹ S. Sheldon,¹ J. Watrous,¹ C. Zoufal,³ A. Dauphin,² and B. Peropadre⁴

¹IBM Quantum, IBM Thomas J. Watson Research Center, USA*

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³IBM Quantum, IBM Research Europe – Zurich, 8803 Rüschlikon, Switzerland

⁴IBM Quantum, IBM Research, Cambridge, MA 02142, USA‡

As quantum computers approach the threshold where they can demonstrably outpace their classical counterparts on certain tasks, the need for a precise, consensus-driven definition of quantum advantage becomes essential. Rapid progress in the field has blurred this term across organizations, architectures, and application domains. Here, we aim to articulate a functional definition for quantum advantage that is both platform-agnostic and empirically verifiable. Building on this framework, we highlight the algorithmic families most likely to achieve early advantage. Finally, we outline our vision for the near future, in which quantum computers enhance existing high-performance computing platforms.

I. INTRODUCTION

Quantum computing has long promised to solve computational problems that are intractable for classical computers [1–7]. Recent advances in quantum hardware, software, and algorithm design have significantly accelerated progress toward this goal. Notably, several experimental demonstrations have shown that quantum processors can now perform accurate computations at scales that surpass the capabilities of brute-force classical simulation of quantum circuits [8, 9]. As these technologies continue to mature, a critical question has emerged at the forefront of the field: When will quantum computers deliver a clear, demonstrable advantage in solving practical, real-world computational tasks? This milestone, commonly referred to as *quantum advantage*, represents a pivotal threshold for the field.

In this work, we propose a functional definition of quantum advantage and delineate the criteria required to substantiate such a claim. Establishing such criteria for when and how quantum advantage is achieved is crucial, not only as a rigorous benchmark of quantum technological progress, but also for identifying application domains with the potential for near-term impact. We also identify important components in this path: methods for the reliable execution of quantum circuits on performant quantum processors, their integration with high-performance computing (HPC) infrastructure, and the development of algorithms that can be executed with high fidelity and yield verifiably correct results.

separation that demonstrably offers superior efficiency, cost-effectiveness, or accuracy than what is attainable with classical computation alone. Although the concept appears straightforward, these criteria remain insufficiently articulated in the current literature.

A. Validation

Establishing trust in the outcome of a computation is one of the foremost tasks in establishing any new computational method. This is particularly critical for a quantum computation, since quantum computers are poised to solve problems that are not efficiently solvable by established classical methods, making a direct comparison challenging. Furthermore, since quantum computing is such a new paradigm, the concern is that simple sanity checks based on reasonable intuition may not be sufficient. Testing on simple cases that can be efficiently evaluated exactly by classical means may not capture the challenges the computation faces when addressing harder instances of the problem. As we argue in the manuscript, there are different ways in which we can ensure that the quantum computation can be validated to produce the correct answer even at a scale where establishing trust in the computation becomes challenging. The most straightforward way is to ensure that the computation itself was performed correctly by providing rigorous error bars, as can be obtained from a fault-tolerant quantum computer, or formally proven error mitigation methods. However, there are also interesting computational problems where the possibility of validation is directly possible by the nature of the answer.

II. DEFINITION OF QUANTUM ADVANTAGE

arXiv:2506.20658v2 [quant-ph] 14 Jul 2025

Definition of quantum advantage

Quantum advantage denotes the *execution of an information processing task on quantum hardware* that satisfies two essential criteria

(1) Separation

It is performed with a quantum separation that *demonstrably* offers superior

A. efficiency,
B. cost-effectiveness, or
C. accuracy

than what is attainable with classical computation alone.

(2) Validation

The correctness of the output can be rigorously validated via:

A. classical verification,
B. scoring and ranking,
C. error bars/ bounds

Elements for quantum advantage

(1) Hardware

Achieving quantum advantage in the near term is fundamentally constrained by the performance of the underlying hardware.

(2) Algorithms

The development of algorithms that can be executed with high fidelity and yield verifiably correct results as a way to bridge the gap between quantum advantage and near-term hardware improvements.

(3) Infrastructure

Realizing near-term quantum advantage—and preparing for fault-tolerant systems anticipated later in this decade—requires extending HPC infrastructure to support modular QPU integration.

(1) Hardware

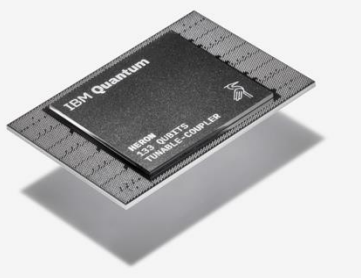
Achieving quantum advantage in the near term is fundamentally constrained by the underlying hardware:

1. The number of qubits
2. The achievable algorithmic depth
3. The fidelity of qubit operations
4. The efficiency of quantum-classical orchestration

IBM's quantum processors:

1. Heron QPU: 156 transmon qubits arranged in modular, scalable architectures (heavy-hex topology).
2. Circuit layer operations per second (CLOPS) above 250,000.
3. Error per layer gate (EPLG) of $2.3\text{E-}3$.
4. Real-time classical controllers enable dynamic circuits and mid-circuit measurement and feedforward.

The fast sampling rates of superconducting qubits makes them particularly attractive for near-term explorations of quantum advantage that rely on a large number of samples.



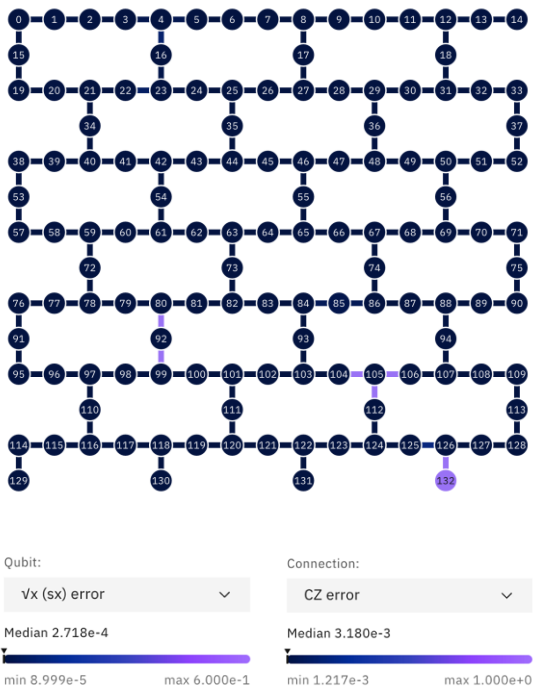
Heron

- R1: 133-qubit systems
- R2: 156-qubit systems
- R3: 156-qubit systems

Tunable coupler architecture

R2: Includes ability to tune away two-level system defects during calibration

IBM Monte Carlo simulation



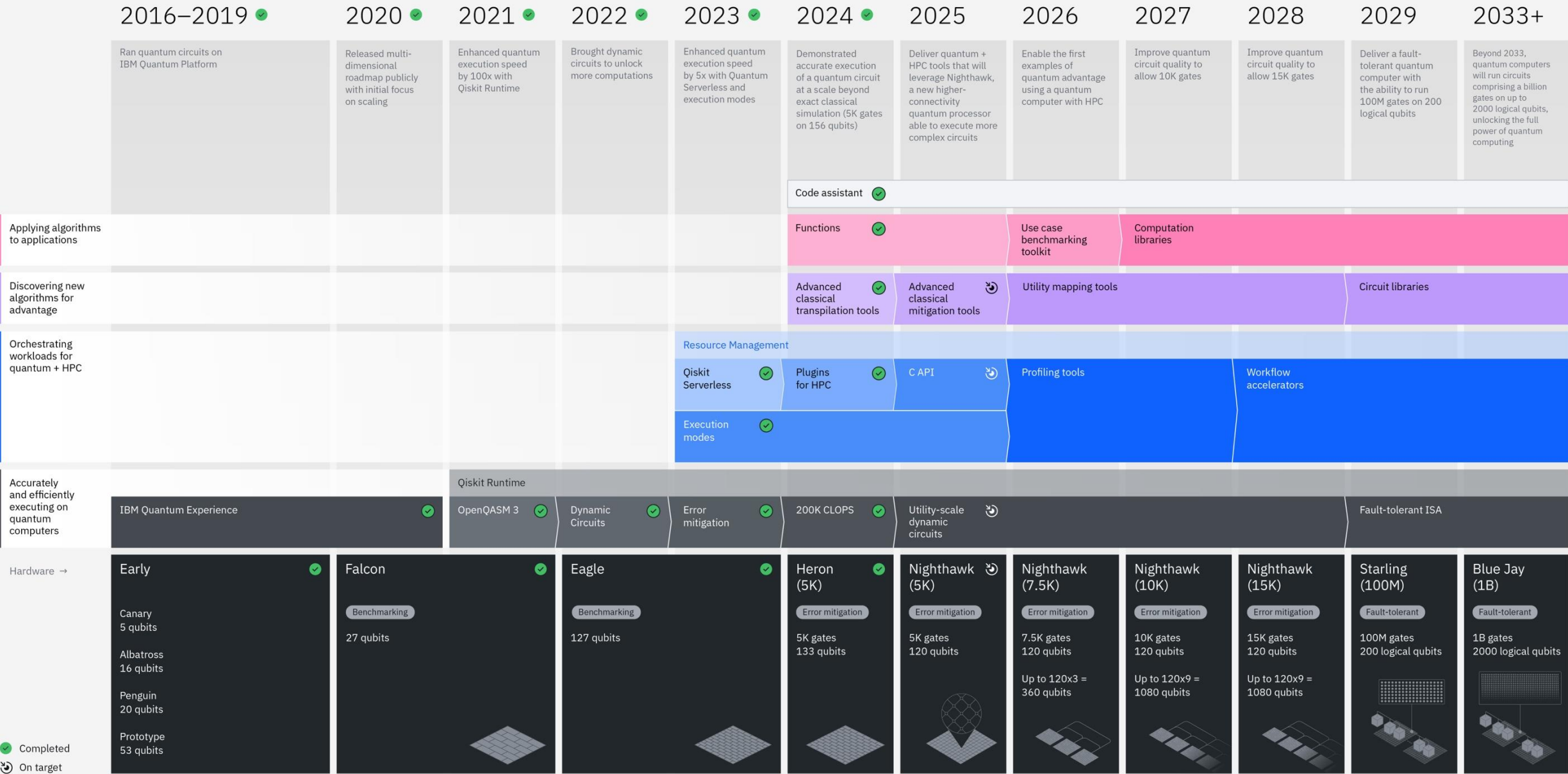
	IBM Sherbrooke Eagle	IBM Monte Carlo (Heron)
Gate error (best system)	0.6%–0.7%	0.3% – Best ~ 0.1%
Crosstalk	High (qubit-qubit collisions)	Almost zero!
Gate time	500–600ns	90–100ns

Iterating on existing innovations

Performance = Scale + Quality* + Speed

Heron R2	156 Qubits	4.4×10^{-3} 2Q Error (4EPLG ₁₀₀ /5)	250k CLOPS
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*Best 2Q gate error 8×10^{-4} as measured by Randomized Benchmarking



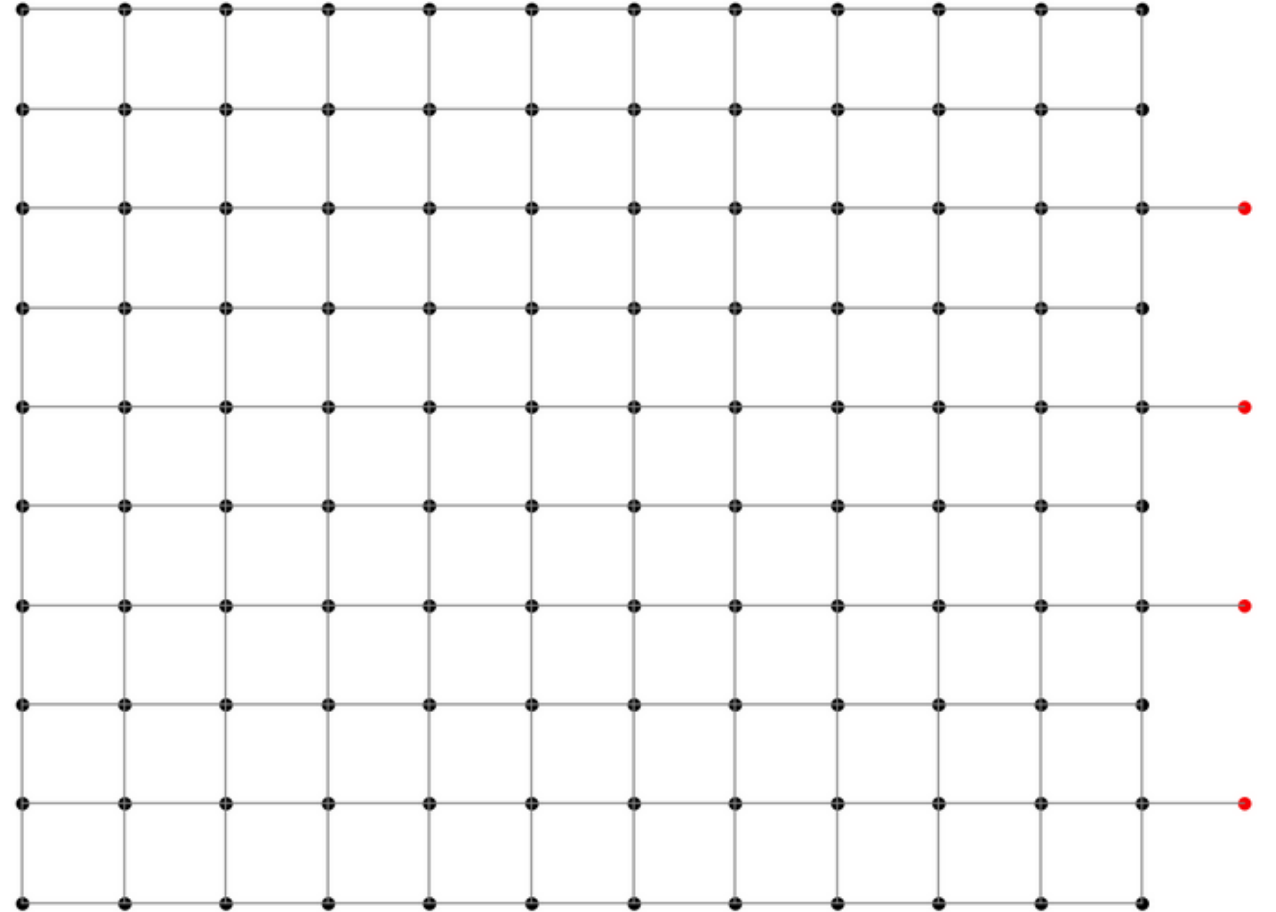
Completed
On target

IBM Quantum Nighthawk

120 qubits

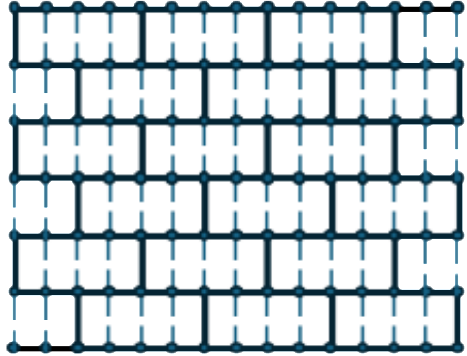
Tunable coupler
architecture in a
square lattice

4-degree connectivity
to enable higher
circuit complexity



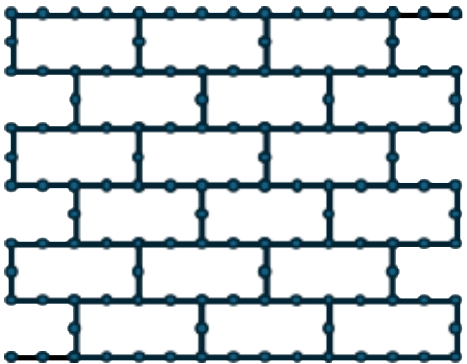
Nighthawk:

The benefits of higher connectivity



Square lattice

4 connections per qubit



Heavy-hex lattice

6 connections per qubit

Circuit depth is directly related to the connectivity between nodes.

For a worst-case circuit, a square lattice has:

→ 5.3x fewer SWAP layers

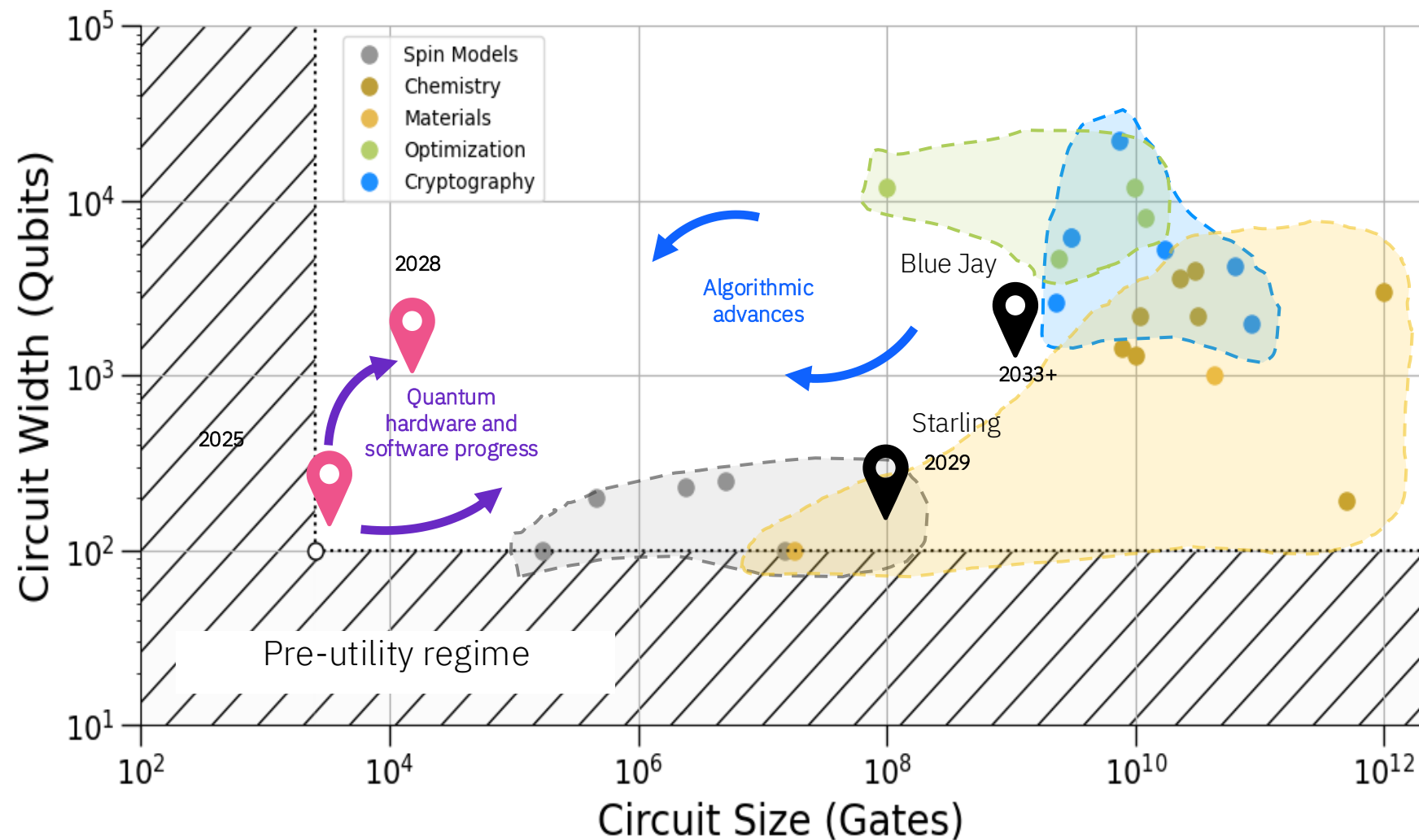
→ 16x fewer CZ layers

Reduced circuit depth enables more complex workloads within the same error budget.

(2) Algorithms

Adoption of quantum computing systems enables **new applications or use cases to be discovered**, potentially with current or near-future systems.

Investments in **algorithm research** should enable known applications to be realized sooner.



Quantum Working Groups

IBM Quantum working groups bring together the best scientists in our field to accelerate our path to achieving Quantum Advantage across domain areas

Optimization

Quantum Optimization: Potential, Challenges, and the Path Forward

Nature Reviews Phys. (2024)
<https://doi.org/10.1038/s42254-024-00770-9>



Quantum Research

↳ [Blog](#)

Recent publications from members of this WG: [exploring the potential for quantum advantage in mathematical optimization](#)

Materials + HPC

Quantum-centric Supercomputing for Materials Science: A Perspective on Challenges and Future Directions

FGCS 160, 666-710 (2024)



High-Energy Physics

Quantum Computing for High-Energy Physics: State of the Art and Challenges. Summary of the QC4HEP Working Group

PRX Quantum 5, 037001 (2024)



Healthcare & Life Sciences

Towards quantum-enabled cell-centric therapeutics

arXiv:2307.05734 [quant-ph]



Sustainability

Collaborative projects in the fields of Materials and Energy leveraging quantum computers.



The Advantage Tracker

Advantage trackers

Track verifiable quantum advantage claims across three pathways, with clear evidence requirements and contributing institutions.

Observable estimations

Variational problems

Classically verifiable problems

Variational problems

Submissions must bound the ground-state energy from above. Verified entries include evidence that the ansatz and optimization respect the variational principle.

View hamiltonians

Open submission ticket

Date	Name / Institutions	Method	Hamiltonian	Qubits	Gates	Energy (Eh) [upper, lower bound]	Runtime (seconds)	Compute resources
2025-11-21	[4Fe-4S] (54e, 36o) CISD By: IBM, RIKEN	Configuration Interaction with singled and doubles	4fe_4s_54e_36o			-326.7421234190204 [N/A, N/A]	Q: - C: -	Q: - C: -
2025-11-21	[4Fe-4S] (54e, 36o) RHF By: IBM, RIKEN	Restricted Hartree-Fock	4fe_4s_54e_36o			-326.54741547585223 [N/A, N/A]	Q: - C: -	Q: - C: -
2025-11-20	TrimCI_202511_6e7 By: University of Wisconsin – Madison	TrimCI	4fe_4s_54e_36o			-327.175 [N/A, N/A]	Q: - C: 50400	Q: - C: 12 Node × 2 × AMD EPYC 7713 64C 2GHz = 1536 cores
2025-11-20	TrimCI_202511_1e4 By: University of Wisconsin – Madison	TrimCI	4fe_4s_54e_36o			-326.913 [N/A, N/A]	Q: - C: 900	Q: - C: Apple M1 Ultra, 20 cores
2025-11-20	TrimCI_202511_1e3 By: University of Wisconsin – Madison	TrimCI	4fe_4s_54e_36o			-326.81 [N/A, N/A]	Q: - C: -	Q: - C: Apple M1 Ultra, 20 cores

<https://quantum-advantage-tracker.github.io/>

(3) Infrastructure

A common misconception frames quantum advantage as a competition between quantum and classical machines. In practice, however, the goal is not to replace classical computing, but to **demonstrate that a hybrid architecture can outperform classical systems alone.**

Classical processors remain essential throughout the quantum workflow, supporting tasks such as system control, calibration, circuit optimization, and post-processing.

Quantum-centric supercomputing (QCSC) envisions QPUs as specialized co-processors tightly integrated with CPUs and GPUs.

Realizing near-term quantum advantage—and preparing for fault-tolerant systems anticipated later in this decade—requires **extending HPC infrastructure to support modular QPU integration (e.g., via SLURM, Qiskit).**

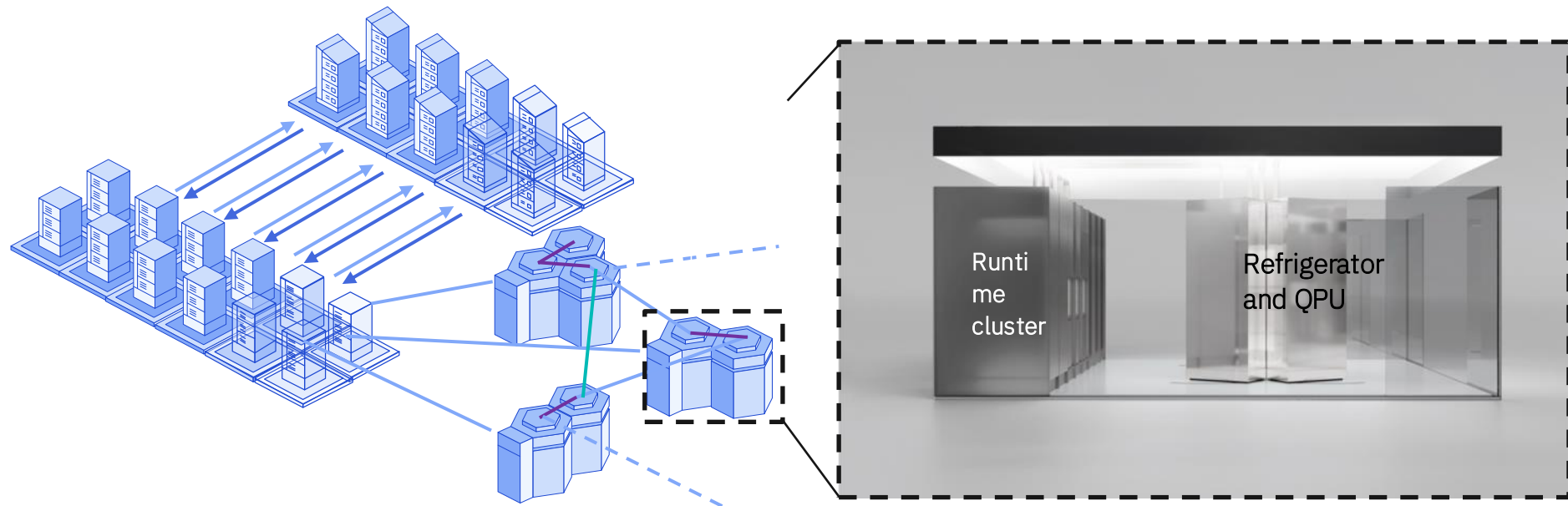
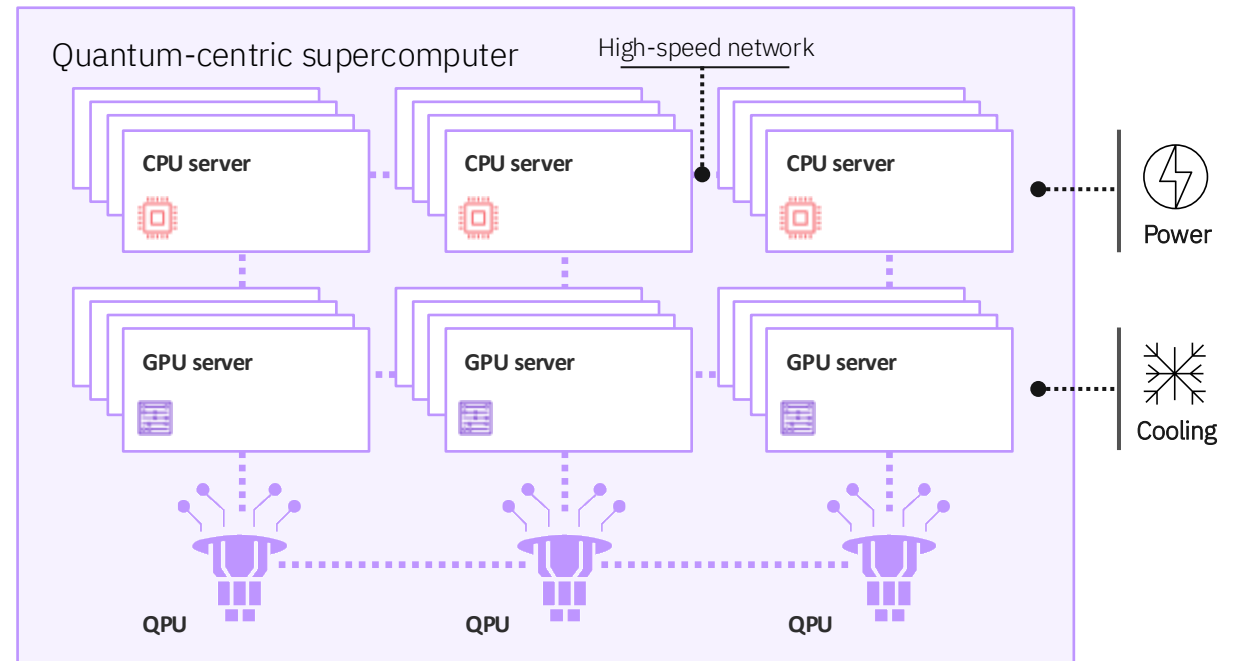
Quantum-centric supercomputing (QCSC)

Delivering impactful quantum computing requires the interplay of quantum and classical resources at scale.

A quantum-centric supercomputer...

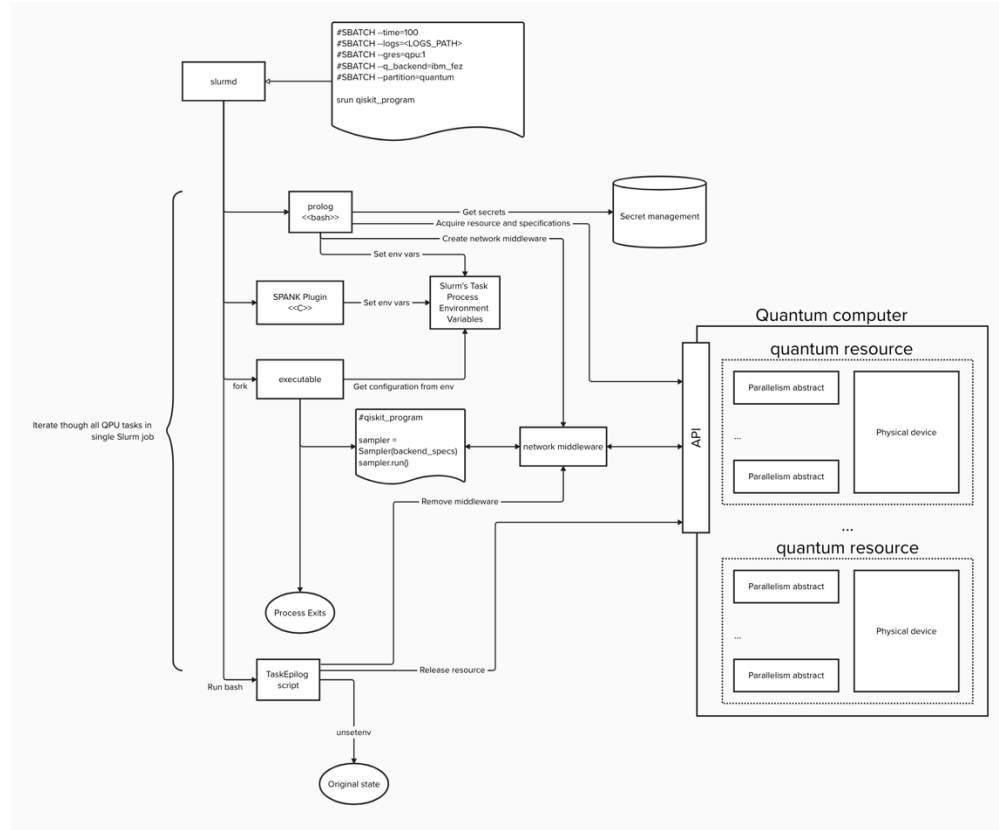
...is not just a larger quantum computer.

...comprises **QPUs**, **GPUs** and **CPUs** required to unlock the next generation of computing.



Quantum resources for workload management systems

Make quantum computers native to any high-performance compute environment



High level architecture



Current state: quantum resources are supported via Slurm plugin interface.

Next step: GRES plugin for Slurm and other workload management systems support (LSF, PBS, etc.)

<https://github.com/qiskit-community/spank-plugins>

<https://arxiv.org/abs/2506.10052>

IBM Quantum



Key milestones in bringing useful quantum computing to the world

2023

Establish quantum utility



2026

Demonstrate quantum advantage



2029

Deliver the first large-scale, fault-tolerant quantum computer



Removing the roadblocks to fault tolerance at scale

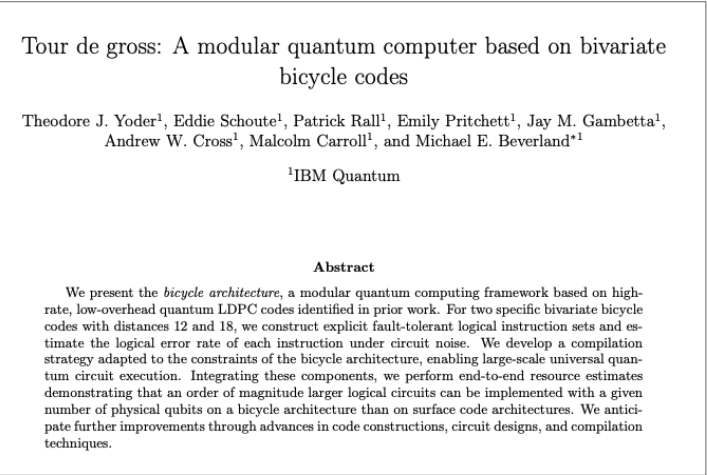
Is there a code that scales efficiently?



Nature 627, 778–782 (2024)

Quantum memory using bivariate bicycle codes requires ~10x fewer physical qubits than the surface code

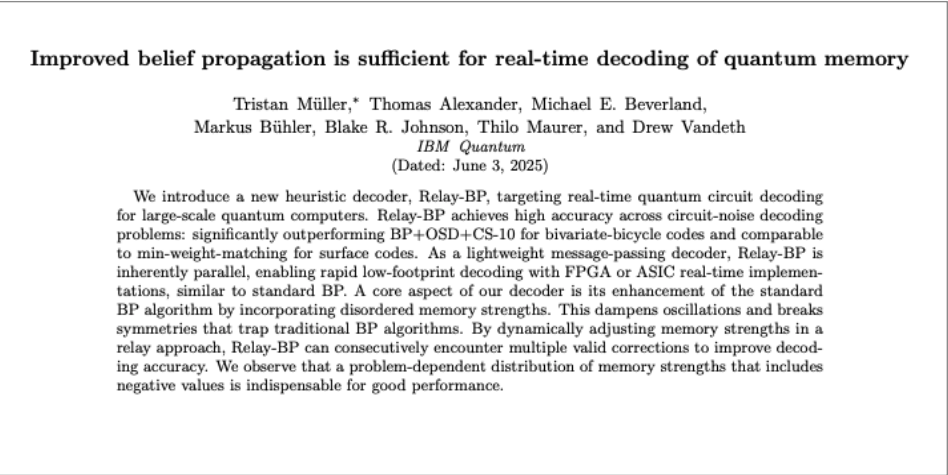
Does this code retain its advantage during computation?



arXiv:2506.03094 (2025)

End-to-end resource estimates demonstrating that bicycle architectures retain an order of magnitude qubit advantage over surface code architectures when implementing large logical circuits

Is real-time decoding possible?



arXiv:2506.01779 (2025)

Demonstrating the feasibility of real-time decoding of qLDPC codes such as bivariate bicycle codes with a modified algorithm (Relay-BP) that eliminates the need for a second-stage decoder.

on

on

2016–2019		2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2033+
Ran quantum circuits on IBM Quantum Platform		Released multi-dimensional roadmap publicly with initial focus on scaling	Enhanced quantum execution speed by 100x with Qiskit Runtime	Brought dynamic circuits to unlock more computations	Enhanced quantum execution speed by 5x with Quantum Serverless and execution modes	Demonstrated accurate execution of a quantum circuit at a scale beyond exact classical simulation (5K gates on 156 qubits)	Deliver quantum + HPC tools that will leverage Nighthawk, a new higher-connectivity quantum processor able to execute more complex circuits	Enable the first examples of quantum advantage using a quantum computer with HPC	Improve quantum circuit quality to allow 10K gates	Improve quantum circuit quality to allow 15K gates	Deliver a fault-tolerant quantum computer with the ability to run 100M gates on 200 logical qubits	Beyond 2033, quantum computers will run circuits comprising a billion gates on up to 2000 logical qubits, unlocking the full power of quantum computing
IBM Quantum Experience	Qiskit	Application modules	Qiskit Runtime	Quantum Serverless	AI-enhanced quantum	HPC-Quantum integration	Advantage candidates	Error correction decoder	Workflow accelerator	Fault-tolerant ISA		
	Open-source SDK for building and compiling circuits for quantum hardware	Modules for domain specific application and algorithm workflows	Performance and abstraction through primitives	Demonstrate concepts of managing quantum and cloud classical compute for an end to end workflow	Prototype demonstrations of AI-enhanced circuit transpilation	Realize an integration of classical HPC and a quantum computer at utility scale	Define problem types for advantage in 2026	Demonstrate a real-time error correction decoder	Demonstrate a workflow accelerator that streamlines execution for a known advantage workflow	Demonstrate a complete instruction set architecture including magic state distillation for FTQC		
Early	Falcon	Hummingbird	Eagle	Osprey	Condor	Flamingo	Loon	Kookaburra	Cockatoo	Starling		
Canary 5 qubits Albatross 16 qubits Penguin 20 qubits Prototype 53 qubits	Demonstrate scaling with I/O routing with bump bonds	Demonstrate scaling with multiplexing readout	Demonstrate scaling with MLW and TSV	Enabling scaling with high density signal delivery	Single-system scaling and fridge capacity	Demonstrate scaling with l-couplers	Demonstrate c-couplers and next-generation packaging for FTQC	Demonstrate a complete module consisting of a logical processing unit and quantum memory	Demonstrate entanglement of modules using a universal adapter	Demonstrate multiple modules and magic state distillation		
		Egret			Heron	Crossbill						
		Tunable coupler demonstration			Architecture based on tunable-couplers	Demonstrate m-couplers						

Completed
 On target

 Completed
 On target

IBM Quantum Starling

The first large-scale
fault-tolerant
quantum computer

Landing 2029

IBM Quantum Starling

200 logical qubits

100 million quantum gates

—

2029

IBM Quantum Data Center
Poughkeepsie, New York

IBM Quantum Blue Jay

2000 logical qubits

1 billion quantum gates

—

2033+

IBM Quantum System Two (4x)

Supports 1000+ physical qubits

15000+ quantum gates

—

2025



But we can't (and don't want) to do this alone...

...to realize Quantum Advantage, we need a dedicated community of industries, researchers, and practitioners to activate the use of quantum algorithms.

IBM Quantum Network

The **largest** quantum ecosystem in the industry

The **IBM Quantum Network** is the core of our global user community and ecosystem — driving innovation and shaping the future of quantum computing.

Universities and research laboratories

Business partners

Startups

Industry leaders

600K+

Registered users worldwide

5K+

Research papers using IBM quantum computers

300+

Organizations in the IBM Quantum Network

1300+

Qiskit Advocates

Ecosystem

IBM Quantum Platform hosts the strongest ecosystem advancing quantum computing

The IBM Quantum Network has 300+ members

Industry members	50+
Commercial partners and startups	55+
Academic and research institutions	170+

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Programming Quantum Computers with Qiskit SDK

Qiskit SDK

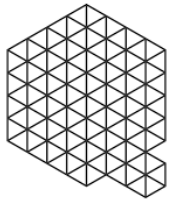
arXiv:2405.08810

The lingua franca of quantum computing; write once and execute quantum circuits on **10+** different hardware providers

IBM Quantum



IQM



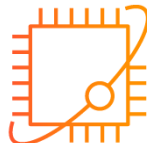
Azure Quantum



ALICE & BOB



IONQ



Amazon Braket
Get started with
quantum computing

Superstaq



QUANTINUUM

rigetti

Quantum SDK Preferred
(2025 Unitary Fund Survey)

69%

Qiskit contributors that
are external to IBM

74%

Dependent Qiskit Projects

7400+

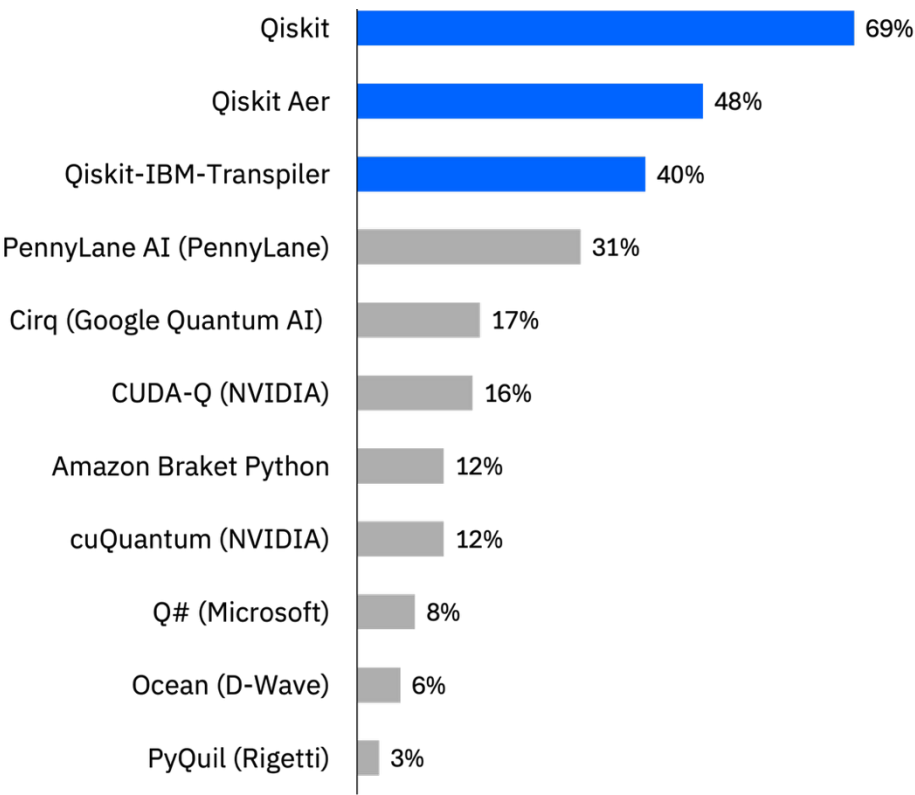
Qiskit is the most preferred and performant quantum SDK

Speed	129x faster in wall clock time (Qiskit 2.0.2 vs. Tket 2.6.0)	83x faster mean transpilation time (Qiskit 2.2.0 vs. Tket 2.6.0)
Quality	29% fewer 2Q gates (Qiskit 2.2.0 vs. Tket 2.6.0)	Only SDK not to fail any tests
Extensibility	5,700+ more dependent projects (Qiskit: 7400+ vs. PennyLane: 1700+)	

Read the Benchpress paper
↳ <https://www.nature.com/articles/s43588-025-00792-y>

Results reported from Benchpress tests of ~1000 circuits (<https://github.com/Qiskit/benchpress>) and published in *Nature Computational Science* (2025): <https://www.nature.com/articles/s43588-025-00792-y>. Timing and quality measured using only completed tests. Dependencies from Github insights. Current as of 12/9/25.

Full-stack development platforms, compilers, and simulators adoption



Source: Unitary Foundation 2025 Quantum Open Source Software Survey

Accelerate application research with a high-performance software stack powered by Qiskit

Qiskit SDK

Preferred by 69% of the developer community

7400+ dependent projects

83x faster mean transpilation time than nearest competitor TKET

Qiskit Runtime

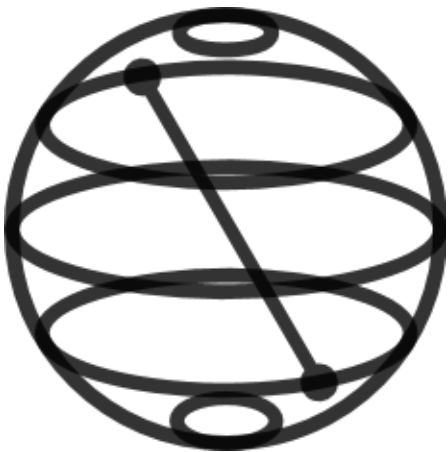
Accurately and efficiently execute estimation and sampling tasks in a near-time environment

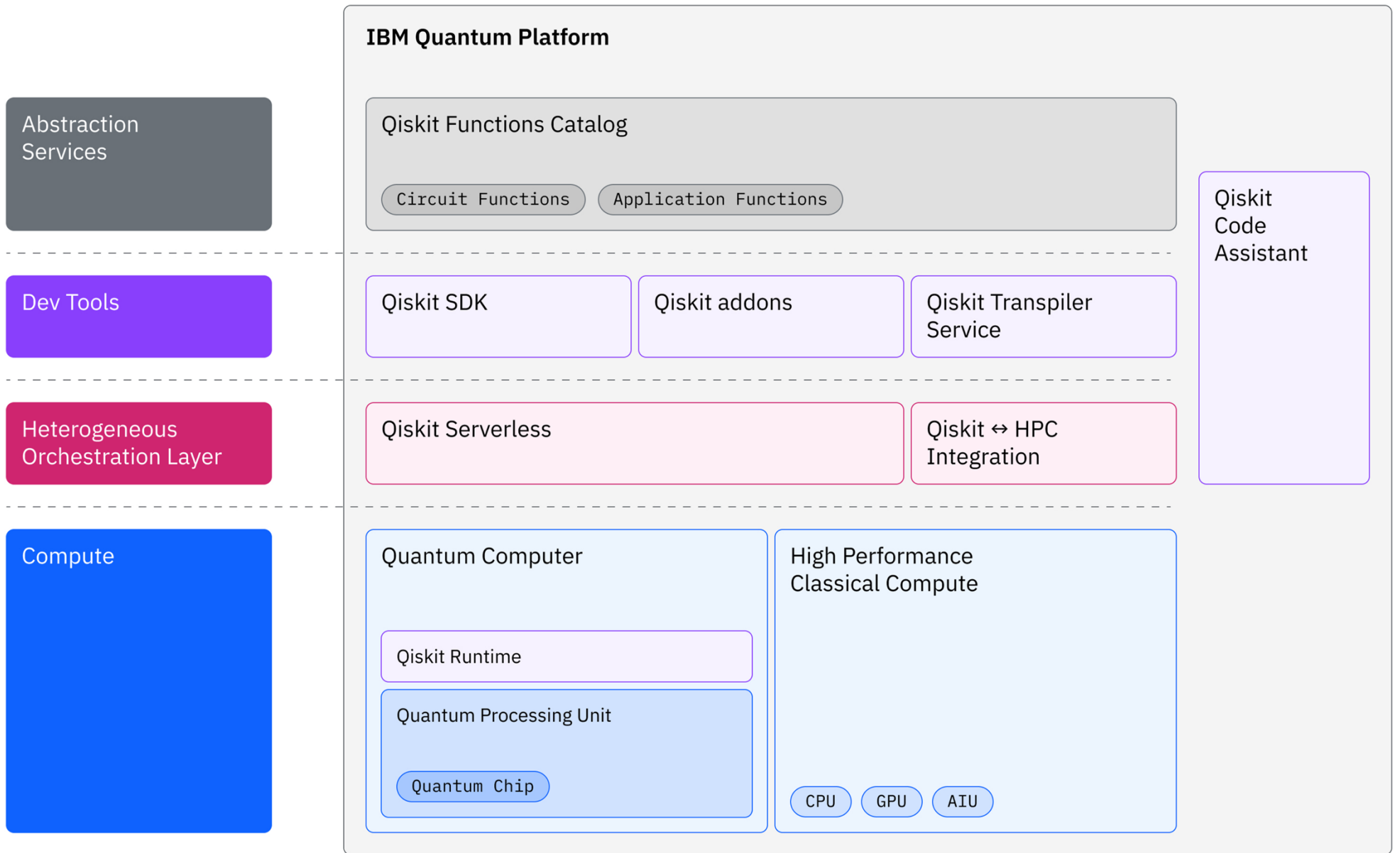
Quality: Up to 5K gates
Speed: 250k+ CLOPS

Qiskit Serverless

Run quantum-centric supercomputing workloads across QPUs, CPUs, and GPUs in the cloud

Simplifies heterogeneous application development for researchers

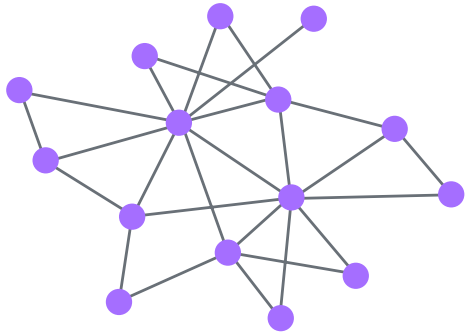




Qiskit patterns

Step 1

Map classical inputs to a quantum problem



Step 2

Optimize problem for quantum execution.

```
PassManager([UnitarySynthesis(),
              BasisTranslator(),
              EnlargeWithAncilla(),
              AIsWap(),
              Collect1qRuns(),
              Optimize1qGates(),
              Collect2qBlocks(),
              ConsolidateBlocks()])
```

Step 3

Execute using Qiskit Runtime Primitives.



Sampler

circuit($\vec{\theta}$)

000101...,
110110...

bit-strings



Estimator

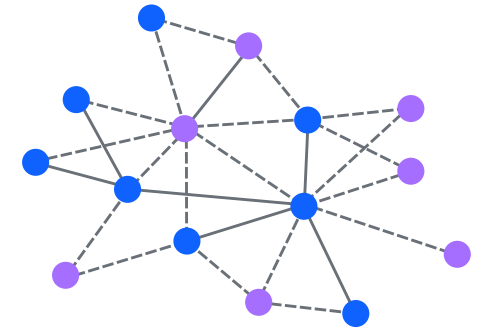
circuit($\vec{\theta}$) +
observable $\hat{O}$

$\langle O \rangle$

expectation
value

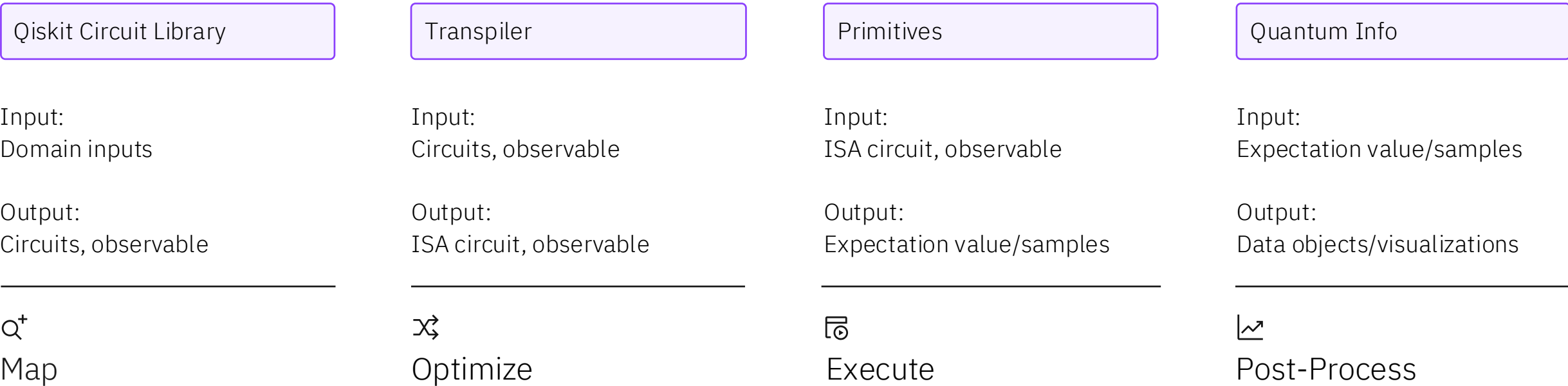
Step 4

Post-process, return result in classical format.



Qiskit SDK sets the foundation

Qiskit SDK gives us a base layer of building blocks for building and running quantum algorithms



Qiskit addons build on the Qiskit SDK

A collection of research capabilities developed as modular tools that can plug into a workflow to design new algorithms at the utility scale

Starting with multi-product formulas (MPF), approximate quantum compilation (AQC-Tensor), operator backpropagation (OBP), and sample-based quantum diagonalization (SQD).

AQC-Tensor

MPF

Qiskit Circuit Library

Input:
Domain inputs

Output:
Circuits, observable

Q⁺
Map

OBP

Circuit cutting

Transpiler

Input:
Circuits, observable

Output:
ISA circuit, observable

⌘
Optimize

Primitives

Input:
ISA circuit, observable

Output:
Expectation value/samples

⌚
Execute

SQD

M3

Quantum Info

Input:
Expectation value/samples

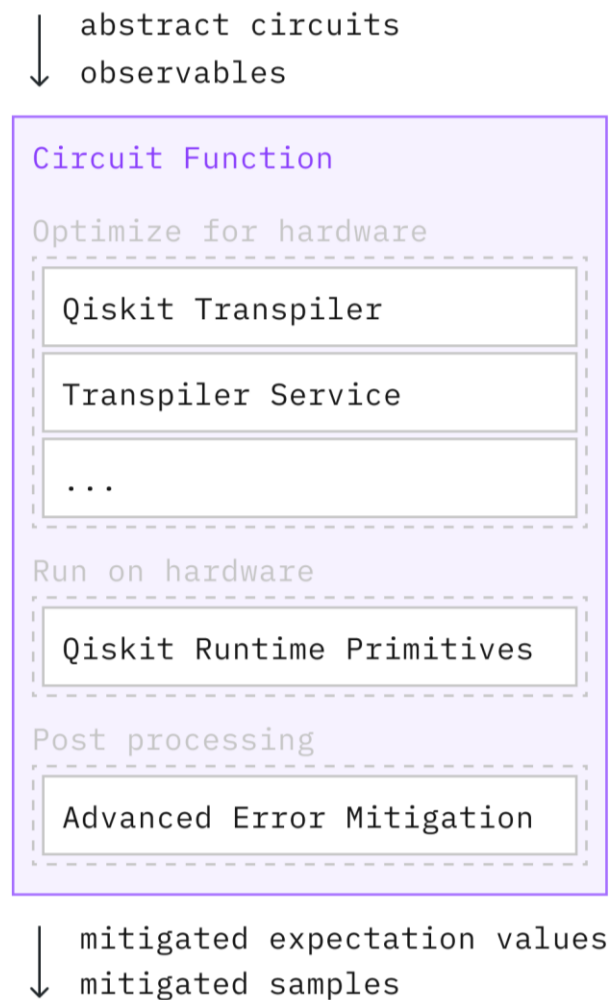
Output:
Data objects/visualizations

📈
Post-Process

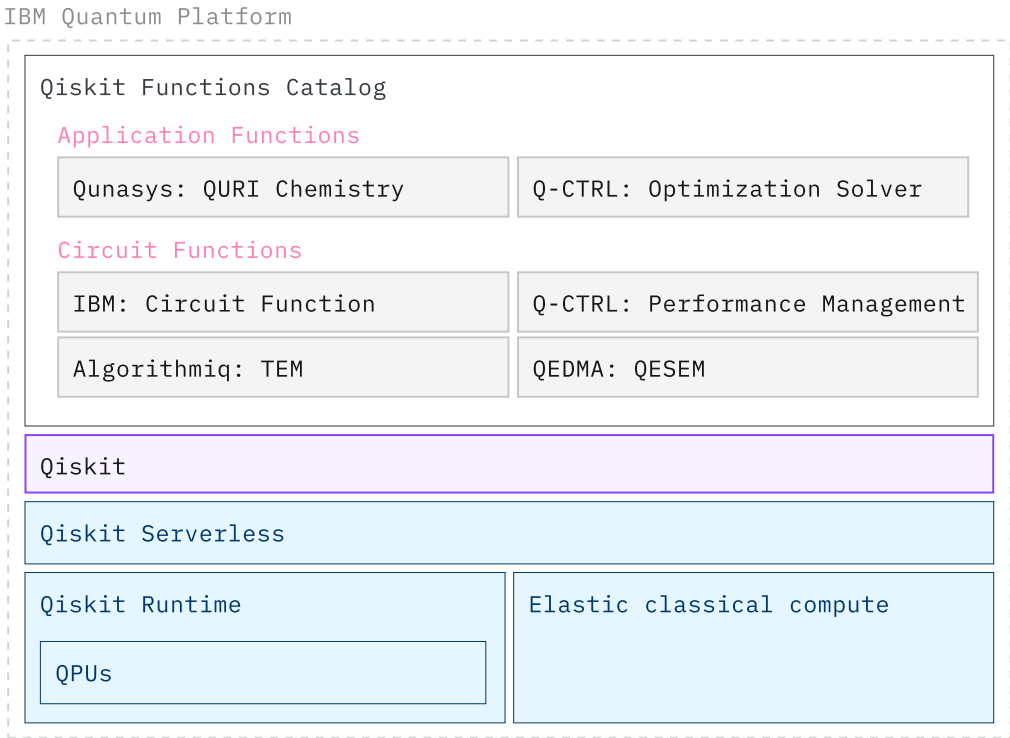
Qiskit Functions are abstracted services, designed to accelerate development

IBM is [partnering](#) with the broader quantum computing [ecosystem](#) to offer a catalog of [managed, utility-scale services](#) to unlock new possibilities:

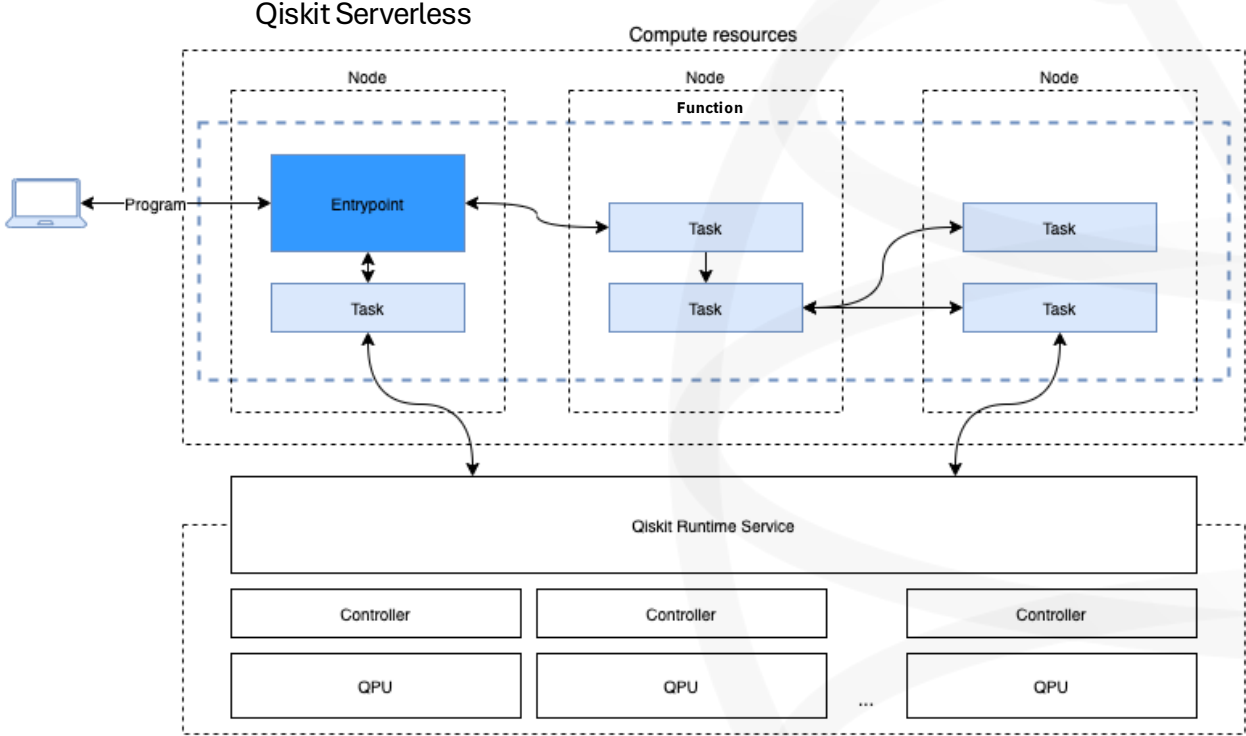
- **Circuit Functions:** Enabling quantum computational scientists to discover new algorithms and applications, without needing to manage transpilation, error suppression, or error mitigation
- **Application Functions:** Enabling data scientists and enterprise developers to integrate quantum into industry workflows, while leveraging familiar domain abstractions



Qiskit Functions run on Qiskit Serverless



Function providers can deliver Qiskit Functions across multi-cloud environments with **Qiskit Serverless**



With **Qiskit Serverless** users can build, deploy, and run workloads remotely using the compute resources of the IBM Quantum platform.

Circuit Functions

Enable quantum computational scientists to discover new algorithms and applications

IBM Quantum Platform

Searchus-eastSign in

Qiskit Functions Catalog

Need help?
View docs →

Looking for which functions you have access to? To check which functions are enabled for your instance, [view docs](#) →

Application functions

Integrate quantum into your application workflows, with problem-specific classical inputs and outputs.

QUICK-PDE

ColibriTD

Solve partial differential equations and complex multi-physics problems with IBM Quantum QPUs.

Quantum Portfolio Optimizer

Global Data Quantum

Experiment optimal investment strategies across dynamic portfolio optimization problems.

HI-VQE Chemistry

Qunova Computing

Solve approximate molecular ground states using an iterative subspace diagonalization, with computational efficiency and accuracy

Iskay Quantum Optimizer

Kipu Quantum

Automated workflow to solve optimization problems up to 156Q, leveraging circuit compression and counterdiabatic protocols.

Singularity Machine Learning

Multiverse Computing

Solve real-world classification problems on quantum hardware without requiring quantum expertise.

Optimization Solver

Q-CTRL

Solve optimization problems at full device size (156Q) from high-level inputs, automatically & efficiently achieving high-accuracy results at utility-scale.

Circuit functions

Simplify application research with pre-built tools to help with transpilation, error mitigation, and suppression.

Circuit function

IBM

Compute accurate expectation values on QPUs, with AI-powered circuit optimization, and advanced error mitigation methods.

Tensor-Network Error Mitigation

Algorithmiq

Tensor-network error mitigation with low shot overhead and reduced runtime usage.

QESEM: error suppression & mitigation

Qedma

Accurate results for large circuit volumes using characterization, transpilation, error suppression and unbiased mitigation.

Performance Management

Q-CTRL

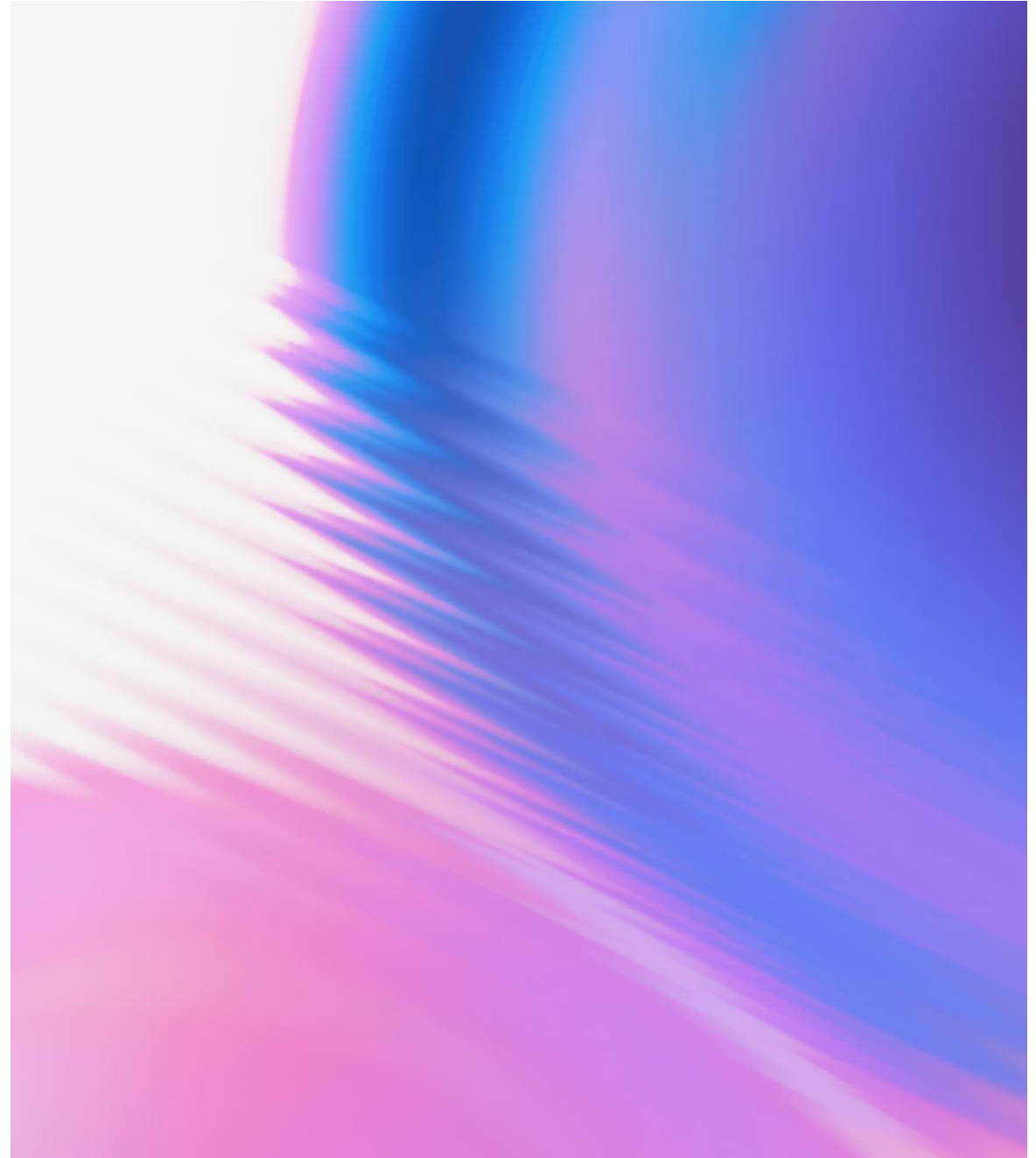
Reduce errors without sampling overhead, with automated, AI-driven error suppression.

TermsPrivacyCookie preferencesSupport

English

IBM Quantum Platform

Access, Learning,
and Documentation



IBM Quantum Platform: getting started

Go to the IBM Quantum Platform

<https://quantum.ibm.com>

and create an IBMid to run your own circuits!

Everyone can get in, you just need an email account!



Get started today on
IBM Quantum Platform

Sign in to IBM Quantum

Continue with IBMid



New to IBM Quantum?

[Create an IBMid](#)

Having trouble signing in?

Try signing in with an IBMid. If you are still having issues, contact the [IBMid help desk](#).

Create an IBM Cloud account

Already have an IBM Cloud account? [Log in](#)

Account information

Email

Enter an email address.

Password

OR

Sign up with Google



Sign up with Red Hat



Next



☐ Verify email

☐ Personal information

☐ Account notice

Create account

IBM Quantum Platform

To bring useful quantum computing to the world, quantum computing must be easy to use.

To support this mission, we are gathering all IBM Quantum resources into one easy-to-access location.

Our goal is to build a mature and stable quantum programming experience.

IBM Quantum Platform

Search2991026 - Quantum Client Enable...us-east

Marcel Pfaffhauser

IBM Quantum Platform

Help us improve Qiskit and IBM Quantum Platform! Share your thoughts in our [annual feedback survey](#)

API key [Create +](#) | [View all](#)

Instances [View all](#)

Educational intern... CRN

Internal | 1h 20m 20s remain

Educational premi... CRN

Premium | 1h 10m 49s remain

My recent workloads [View all](#)

ID	Status	Instance	Created	QPU	Usage

QPUs [View all](#)

12

ibm_pittsburgh 156q | Heron r3

ibm_kingston 156q | Heron r2

ibm_fez 156q | Heron r2

Featured resources

Documentation

Hit the ground running with Qiskit, our open-source toolkit for quantum computing.

[View the docs](#)

Tutorials

Explore utility-grade examples that leverage IBM Quantum and Qiskit technologies.

[Browse tutorials](#)

Composer

Build, simulate, and run quantum circuits with a drag-and-drop interface.

[Start building](#)

Announcements

Stay up to date with the latest news, service alerts, and product updates.

[View announcements](#)

R2P Program | © 2026 IBM Corporation

IBM Quantum Platform

Run your own quantum circuits on IBM Quantum systems!



QPU's you do not have access to with any instance appear with a lock icon below.

Search by QPU name

All QPU's (12) |

↑↓

🔍

ibm_marrakesh QPU status Online Processor type Heron r2 Qubits 156 2Q error (best/layered) 1.23e-3/3.27e-3 CLOPS 195K	ibm_fez QPU status Online Processor type Heron r2 Qubits 156 2Q error (best/layered) 2.07e-3/4.70e-3 CLOPS 195K	ibm_torino QPU status Online Processor type Heron r1 Qubits 133 2Q error (best/layered) 1.83e-3/7.72e-3 CLOPS 210K	ibm_kyiv QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.08e-3/1.43e-2 CLOPS 30K
ibm_sherbrooke QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.59e-3/1.75e-2 CLOPS 30K	ibm_brisbane QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.22e-3/1.77e-2 CLOPS 180K	test_eagle_us-east QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) N/A CLOPS N/A	🔒 ibm_quebec QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.18e-3/1.87e-2 CLOPS 32K
🔒 ibm_renselaer QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.67e-3/1.98e-2 CLOPS 32K	🔒 ibm_kawasaki QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.92e-3/2.57e-2 CLOPS 29K	ibm_strasbourg QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.64e-3/2.67e-2 CLOPS 220K	ibm_brussels QPU status Online Processor type Eagle r3 Qubits 127 2Q error (best/layered) 3.45e-3/3.18e-2 CLOPS 220K

Quantum learning

Kickstart your quantum learning journey with a selection of courses designed to help you learn the basics or explore more focused topics. If you're an instructor, explore content specifically tailored to incorporating quantum in the classroom.



Foundations

Courses to learn about quantum information and how quantum computing works, from the basics onward.

Quantum information and computation I

Basics of quantum information

Learn about quantum information, from states and measurements to quantum circuits and entanglement.

Course

Quantum information and computation II

Fundamentals of quantum algorithms

Learn how quantum algorithms beat classical algorithms for problems including integer factoring and search.

Course

Quantum information and computation IV

Foundations of quantum error correction

Learn how quantum computations can be protected against noise through quantum error correcting codes and fault tolerance.

Course New lesson

Quantum computing in practice

Learn potential use cases and best practices for experimenting with quantum processors having 100+ qubits.

Course New lesson

Focused topics

Continue your learning journey by diving into more focused topics related to quantum computing.

Quantum machine learning

Learn to leverage the power of quantum computing in machine learning methods.

Course New

Variational algorithm design

An overview of variational algorithms: hybrid classical quantum algorithms.

Course

Quantum diagonalization algorithms

Multiple quantum approaches to matrix diagonalization are explored, including VQE, QKD, SKD, and variations of these.

Course New

Utility-scale quantum computing

A collection of learning assets from a 14-lesson course on utility-scale quantum computing.

Course

Quantum computing for business

Explore the business applications of quantum computing and what it takes build a successful quantum initiative.

Quantum business foundations

Builds foundations of quantum computing, technology, and business applications.

Course

Practical introduction to quantum-safe cryptography

Learn the risks that quantum computing poses to cryptography, and explore which approaches are quantum-safe.

Course

Qiskit in classrooms

Explore a suite of instructional materials designed to incorporate quantum computing into traditional STEM courses.

Qiskit in classrooms

Quantum mechanics

Use quantum computers to explore quantum phenomena like superposition and uncertainty.

Module New

Qiskit in classrooms

Computer science

Run some famous quantum algorithms, including Deutsch-Jozsa and QKD.

Module New

IBM Quantum Documentation



IBM Quantum Platform

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Guides

Tutorials

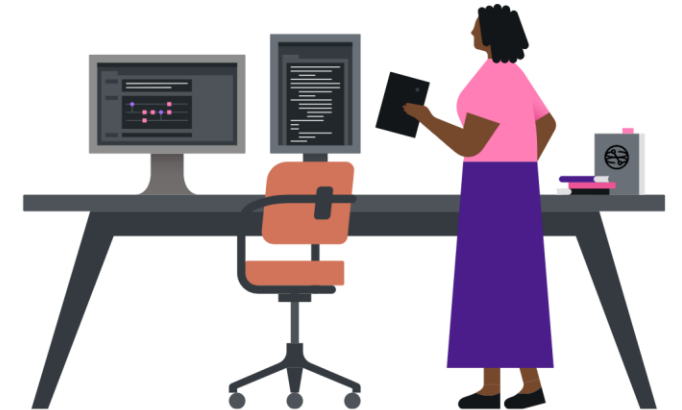
API reference

Additional resources

Documentation

Whether you are ready to code your first circuit or execute a large research workload, you can find documentation for using Qiskit and IBM Quantum hardware at the following links.

Get started with Hello World →



Development workflow

A framework for breaking down domain-specific problems and software capabilities in stages



1 Map the problem to quantum circuits

[Qiskit SDK circuit library](#)

[Construct circuits](#)

[Qiskit SDK's quantum_info module](#)

[See all](#) →



2 Optimize for target hardware

[Transpile with pass managers](#)

[Write a custom transpiler pass](#)

[AI-enhanced transpiler passes](#)

[See all](#) →



3 Execute on target hardware

[Introduction to primitives](#)

[Specify Qiskit Runtime options](#)

[Introduction to execution modes](#)

[See all](#) →

IBM Quantum Tutorials

Improve your coding skills
and learn about quantum
computing with Qiskit



IBM Quantum Platform

Early access

Documentation

Guides

Tutorials

API reference ▾

Additional resources ▾

Tutorials

Overview

Explore workflows toward advantage

Verifiable sampling algorithms New ▾

Observable estimation New ▾

Fault-tolerant algorithms ▾

Leverage Qiskit capabilities

Workload optimization ▾

Addons ▾

Error mitigation ▾

Error detection ▾

Tutorials

Use these tutorials to learn how to apply Qiskit to common quantum computing use cases.

- The section on workflows toward advantage contains end-to-end examples of using a quantum computer to solve real-world problems. These tutorials focus on algorithms that are promising candidates for achieving computational advantage by a quantum computer over a classical computer.
- The section on Qiskit capabilities contains examples that use the latest and most advanced techniques in the Qiskit ecosystem to improve part or the whole of a particular workflow.

Explore workflows toward advantage

Tutorials in this section cover large scale demonstrations of quantum algorithms.

- **Verifiable sampling algorithms**
- **Observable estimation**
- **Fault tolerant algorithms**

Qiskit Youtube

Engaging lectures, tips & tricks, tutorials, community updates and access to exclusive Qiskit content.

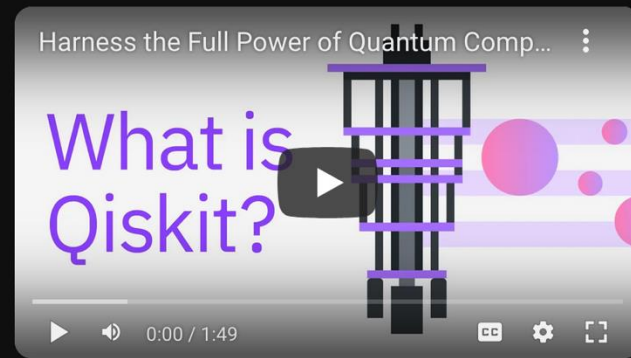
Notable Playlists:

[Coding with Qiskit 1.x](#)

[Quantum Crosstalk](#)

[Qiskit Quantum Seminars](#)

[Quantum Computing in Practice](#)



Harness the Full Power of Quantum Computers

5,061 views • 2 weeks ago

Qiskit is a high-performance software stack built to help developers and researchers harness the full power of quantum computers at the utility scale and beyond.

Get started with Qiskit today

<https://learning.quantum.ibm.com/lear...>

Coding With Qiskit - 1.x ▶ Play all

Welcome back to Coding with Qiskit! Join research scientist Dr. Derek Wang as he walks you through the exciting capabilities of Qiskit 1 for utility scale quantum computing. He'll show you ho...



Coding with Qiskit 1.x Series Announcement
Qiskit ✓
26K views • 10 months ago



EP 1 Introduction to Qiskit
Qiskit ✓
30K views • 10 months ago



EP 2 How to Install Qiskit
Qiskit ✓
73K views • 10 months ago



EP 3 Hello World
Qiskit ✓
64K views • 9 months ago



EP 4 Primitives
Qiskit ✓
17K views • 9 months ago



EP 5 Dynamic Circuits
Qiskit ✓
7.9K views • 9 months ago

Quantum Crosstalk ▶ Play all

The short form show where we sit down with influential members of IBM's quantum team and discuss the latest happenings, events, and product releases.



Qiskit Performance with Dr. Blake Johnson | Quantum...
Qiskit ✓
2.1K views • 5 months ago



What It's Like to Intern at IBM | Quantum...
Qiskit ✓
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AI Enhanced Quantum Software: Quantum Crosstalk...
Qiskit ✓
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Runtime Rundown: Quantum Crosstalk with Jessie Yu
Qiskit ✓
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Quantum for Computational Scientists: Quantum Crosstalk...
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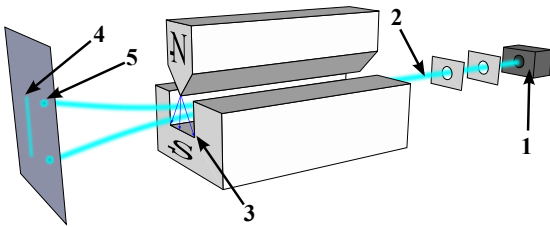


What's Coming to Quantum in 2024: Quantum Crosstalk...
Qiskit ✓
5.8K views • 11 months ago

Modules for Intro Quantum Mechanics

Stern-Gerlach Experiment – a module on spin quantization

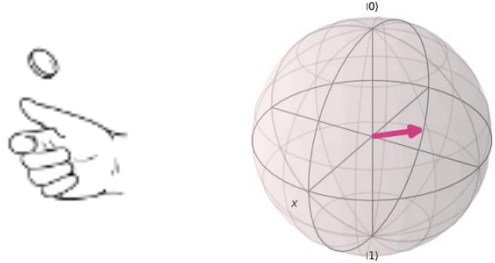
Module 01



- Uses real qubits to reproduce original Stern-Gerlach results and extensions

Superposition – a module on superposition and interference

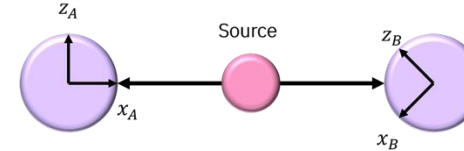
Module 02



- Compares measurement of a qubit in equal superposition to a coin flip and explores limits of that analogy.

Bell's Inequality – a module on entanglement and disproving local realism

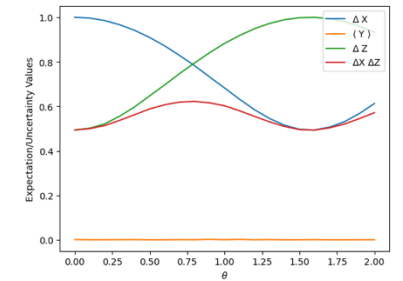
Module 03



- Shows how quantum mechanics violates Bell's inequality using a real quantum computer.

Quantum Uncertainty – a module on uncertainty relations

Module 04



- Checks the generalized uncertainty relations of Pauli observables with a qubit

Modules for Computer Science

Quantum Teleportation

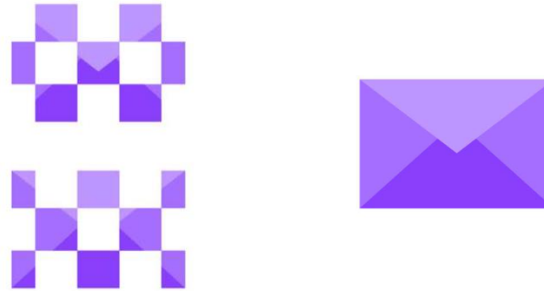
Module 05



- Walks through the quantum teleportation protocol, then shows how to implement it on an IBM quantum computer.

Quantum Key Distribution and BB84

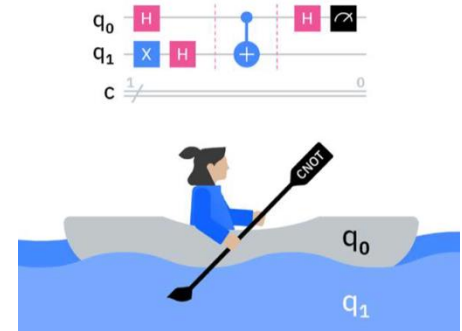
Module 06



- After a primer on cryptography, shows students how the BB84 protocol protects against unnoticed eavesdroppers.

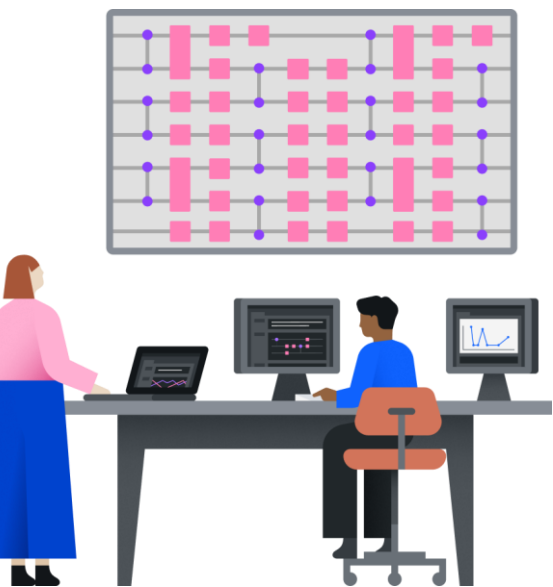
Deutsch and Deutsch-Jozsa algorithms

Module 07



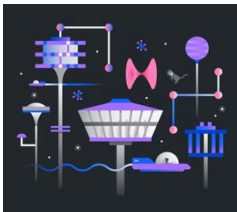
- Shows how the D and D-J algorithms work, with discussions of quantum parallelism and phase-kickback
- Applies the D-J algorithm to the Bernstein-Vazirani problem

IBM Quantum Community



Qiskit Fall Fest

Student-led quantum computing events on campus
↳ Crafted by student leaders, supported by IBM



IBM Quantum Challenge

An annual coding challenge focused on teaching the world how quantum computational scientists use Qiskit.



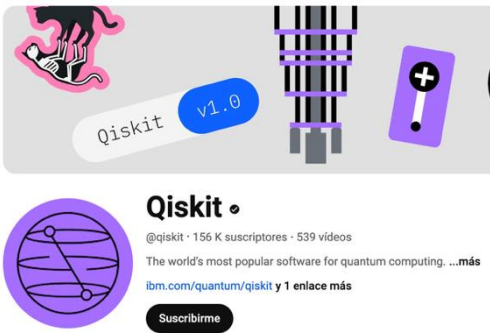
Qiskit Global Summer School

An annual event featuring online lectures delivered by various IBM Quantum experts, as well as live Q&A sessions.



Qiskit YouTube Channel

The Qiskit YouTube channel hosts hundreds of useful videos on quantum computing.



Qiskit Developer Certification

The world's first ever developer certification for programming a quantum computer, setting the benchmark for quantum developer skills.



Qiskit Advocates

A global program that provides support to individuals who actively contribute to the Qiskit community.



What is the Qiskit advocate program?

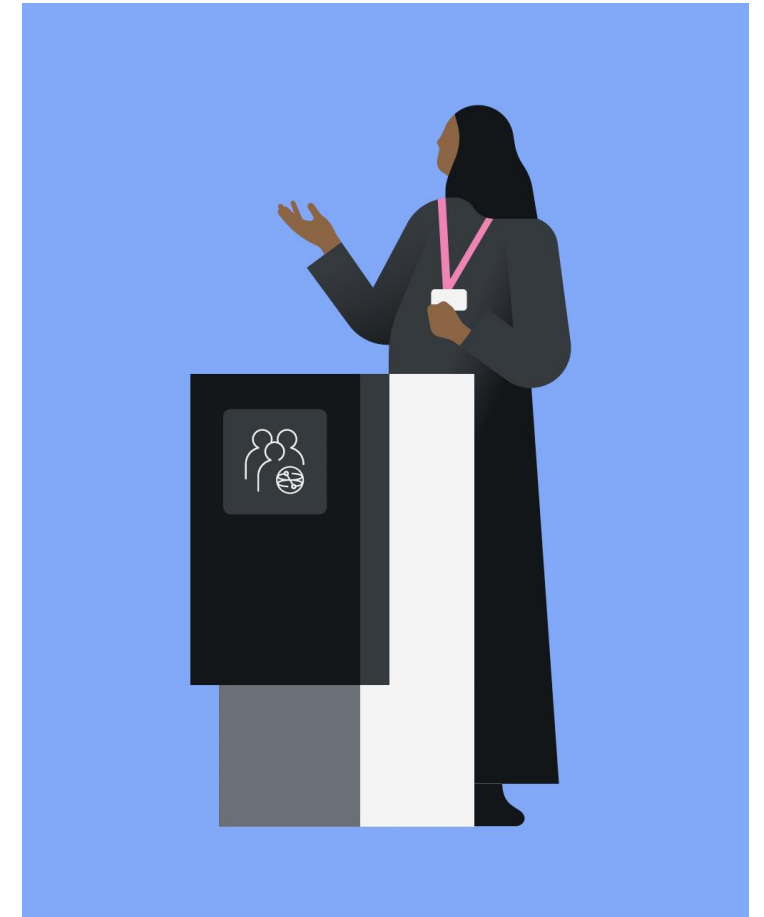
Join 1300+ Qiskit advocates across 70+ countries in advancing quantum computing education and community.

The Qiskit advocate program is an external, global initiative that empowers and enables aspiring leaders in the Qiskit open-source community.

Qiskit advocates enhance their professional growth by:

- Building meaningful connections with quantum experts and peers
- Gaining exclusive access to Qiskit education and events
- Getting unique opportunities to contribute to quantum computing advancement
- Making open-source contributions and sharing knowledge through mentorship
- Participating in Qiskit events, networking, and quantum research and so much more.

Perfect for: Students, software developers, quantum researchers, educators, and anyone looking to upskill and advance their career in quantum computing.



What do Qiskit advocates do?

Activities:

- Give talks and presentations on Qiskit at events and conferences
- Contribute to the Qiskit SDK and Qiskit Ecosystem projects
- Mentor participants in flagship events and host events in your local community
- Provide feedback on Qiskit and IBM Quantum tools as active users
- Collaborate with advocates globally
- Earn points through community contributions to advance through program tiers



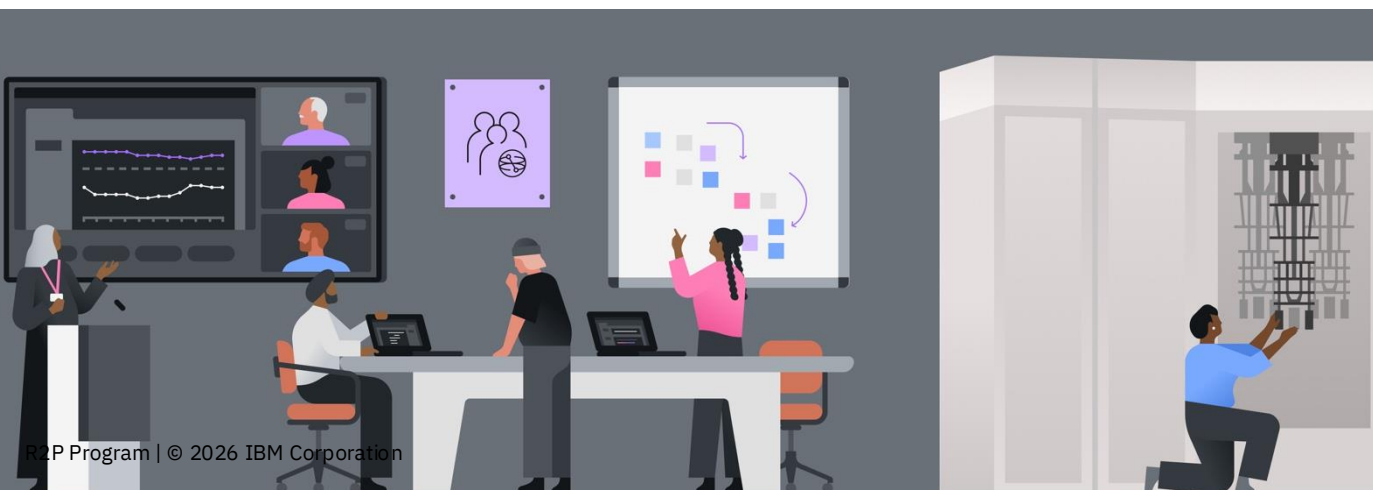
For more information, see: ibm.com/quantum/qiskit#advocates

Ready to get started?

Apply to the program here:



ibm.biz/advocates-application



The Qiskit Advocate Mentorship Program (QAMP) Open House

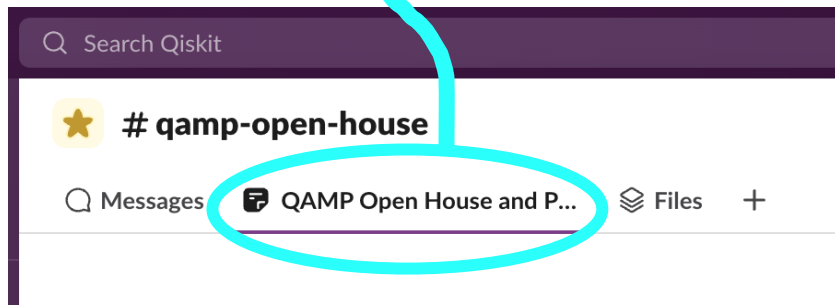
QAMP is an annual 3-month mentorship initiative in the Qiskit advocate program designed to support growth and collaboration within the vibrant quantum community.

For the past three months, Qiskit advocates have been working on quantum computing projects under the guidance of experienced mentors. Many teams are continuing their work beyond the program, with several planning to publish papers and expand their projects further.

We invite you to drop into any (or all) of the sessions linked below to see the minimum viable products (MVPs) these teams have built!

Join **#qamp-open-house** on Qiskit Slack, access the canvas at the top of the channel for session links and agenda, and post questions in project threads!

Agenda



You're invited:



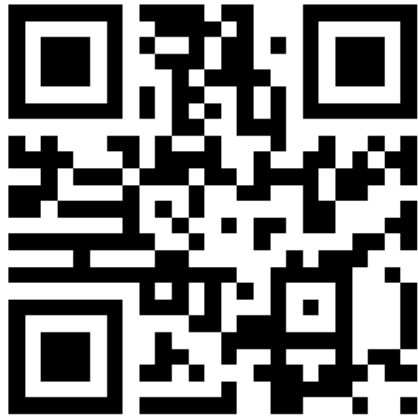
Qiskit Advocate Mentorship Program Open House

- Advocates present their work
- 50 teams | 3-month development journey
- Date: 3 & 4 February



Qiskit Developer Certification

The Qiskit Developer Certification is the world's first ever developer certification for programming a quantum computer. There are over 1,100+ certified Qiskit developers across the globe. This certificate sets a benchmark for the industry in identifying those with quantum developer skills.



Let's create
the future of
quantum
computing
together

fas@zurich.ibm.com

Get started today on
IBM Quantum Platform

